Tools of Artificial Intelligence

Classification Module

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Classification pipeline
Data cleaning
Feature extraction
Feature reduction techniques
Support vector machine (SVM)

Outline

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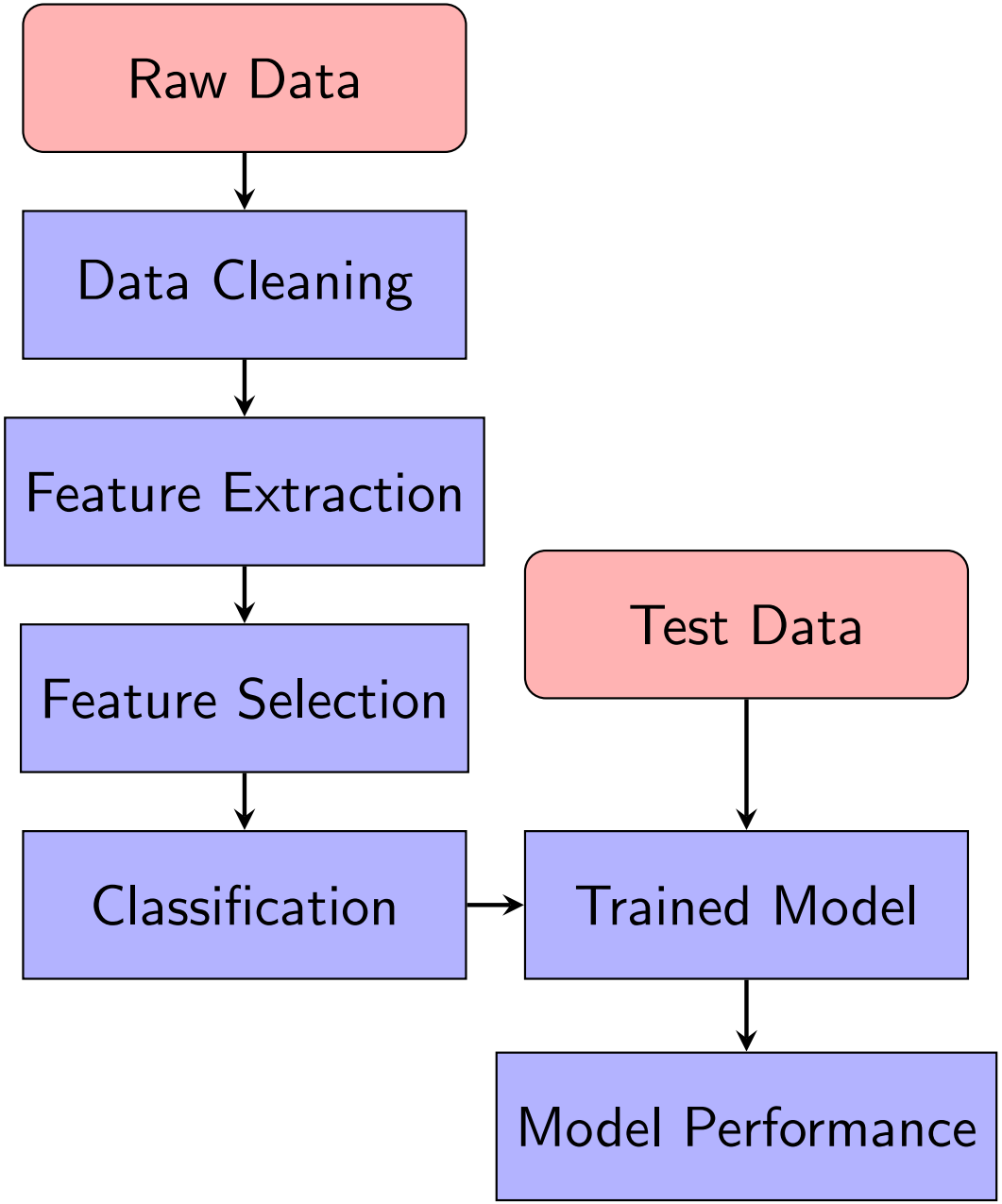
Machine Learning (Classification) Pipeline



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Importance of Data Cleaning and Preprocessing

- **Improves Data Quality**: Ensures the dataset is accurate, consistent, and usable.
- **Prevents Bias**: Removes or corrects errors and inconsistencies that could lead to misleading results.
- **Enhances Model Performance**: Clean data improves the performance of machine learning models by providing them with high-quality inputs.
- **Saves Resources**: Properly preprocessed data ensures that time and resources are not wasted on irrelevant or noisy data.
- **Reduces Overfitting**: Proper preprocessing techniques help to avoid overfitting by removing irrelevant features or noisy data points.
- **Improves Generalization**: Well-prepared data enables the model to generalize better to unseen data.

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Techniques for Data Cleaning and Preprocessing

- Handling Missing Data:
  - Imputation (mean, median, mode, or advanced methods like k-NN).
  - Deleting rows or columns with excessive missing values.
  - Using interpolation for time series data.
- Handling Outliers:
  - Identifying outliers using statistical tests or visualization (e.g., box plots).
  - Removing or correcting outliers based on domain knowledge.
- Data Transformation:
  - Scaling (e.g., min-max scaling, z-score normalization).
  - Encoding categorical variables (e.g., one-hot encoding, label encoding).
- Feature Engineering:
  - Creating new features from existing data to better represent patterns.
  - Feature selection and extraction techniques.
- Filtering:
  - Using filters to remove noise or irrelevant data from the dataset.
  - Techniques like moving averages, Gaussian filters, or Fourier transforms for denoising or smoothing.

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Time Series Data Preprocessing Techniques

- Handling Missing Data:
  - Interpolation (linear, cubic, etc.).
  - Forward or backward filling.
- Trend and Seasonality Decomposition:
  - Use techniques like STL (Seasonal and Trend decomposition using Loess) to remove trends and seasonality.
- Scaling and Normalization:
  - Min-max scaling or z-score normalization of time series data to ensure uniformity.
- Filtering:
  - Apply filters to remove high-frequency noise in time series data.
  - Common filters include Gaussian smoothing, Kalman filters, and moving averages.

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Image Data Preprocessing Techniques

- Resizing:
  - Standardize the size of images to ensure consistency for model input.
- Normalization:
  - Scale pixel values to a range (e.g., [0, 1] or [-1, 1]).
- Data Augmentation:
  - Techniques like rotation, flipping, cropping, and scaling to generate more training data.
- Noise Removal:
  - Apply filters (e.g., Gaussian blur) to remove noise from images.
- Color Space Conversion:
  - Convert to different color spaces (e.g., RGB to grayscale or HSV) to improve model performance.
- Edge Detection:
  - Use techniques like Sobel or Canny edge detection to highlight significant features.
- Filtering:
  - Apply filters like Gaussian blur, median filter, or bilateral filters to smooth the image and reduce noise.

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Challenges in Data Cleaning and Preprocessing

- **Handling Imbalanced Data:** In many cases, data may be imbalanced, which can affect the performance of the model.
- **Noise in Data**: Irrelevant or incorrect data points can negatively affect the performance of machine learning algorithms.
- **Data Leakage**: Leakage of information from outside the training dataset can occur, which leads to overly optimistic model performance.
- **Computational Complexity**: Some preprocessing techniques, especially in large datasets, may be computationally expensive and time-consuming.
- **Preserving Data Distribution**: Preprocessing should not distort the original data distribution, especially in imbalanced classes.

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Preprocessing using filtering

```python
from scipy.signal import butter, filtfilt, iirnotch

def bandpass_filter(data, lowcut, highcut, fs, order=7):
        nyquist = 0.5 * fs
        low = lowcut / nyquist
        high = highcut / nyquist
        b, a = butter(order, [low, high], btype='band')
        y = filtfilt(b, a, data)
        return y

def notch_filter(data, notch_freq, fs, quality_factor=20):
        nyquist = 0.5 * fs
        notch = notch_freq / nyquist
        b, a = iirnotch(notch, quality_factor)
        y = filtfilt(b, a, data)
        return y

fs = 100  # sampling frequency in Hz
# Apply notch filter to suppress 50 Hz signal
filtered_RF = notch_filter(RF, 50, fs)
filtered_RH = notch_filter(RH, 50, fs)

# Apply bandpass filter between 0.5-30 Hz
filtered_signal_RF = bandpass_filter(filtered_RF, 0.5, 30, fs)
filtered_signal_RH = bandpass_filter(filtered_RH, 0.5, 30, fs)
```


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Importance of Features in Machine Learning

- Features are the input variables used by machine learning models to make predictions.
- Good features can significantly improve the performance of a model.
- Feature extraction helps in reducing dimensionality and improving computational efficiency.

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Definition of Features

- A feature is an individual measurable property or characteristic of a phenomenon being observed.
- Features are also known as attributes, variables, or predictors.

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Types of Features

- Statistical Features
- Texture Features
- Entropy Features
- Image Features
- Frequency Domain Features
- Time Domain Features
- Other Features

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Statistical Features

- Mean: $\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$, where N is the number of samples and x_i is the i-th sample.
- Median: The middle value of the sorted samples.
- Mode: The most frequently occurring value in the samples.
- Standard Deviation: $\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$, where μ is the mean.
- Variance: $\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$, where μ is the mean.
- Skewness: $\gamma = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{x_i-\mu}{\sigma}\right)^3$, where σ is the standard deviation.
- Kurtosis: $\kappa = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{x_i-\mu}{\sigma}\right)^4 - 3$, where σ is the standard deviation.

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Related library

```
# Import necessary libraries
import numpy as np
import pandas as pd
import scipy.stats as stats
import scipy.signal as signal
import pywt
import antropy as ant
import matplotlib.pyplot as plt
```

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Statistical features example

```
# Statistical Features
def extract_statistical_features(signal):
        mean = np.mean(signal)
        median = np.median(signal)
        std_dev = np.std(signal)
        variance = np.var(signal)
        skewness = stats.skew(signal)
        kurtosis = stats.kurtosis(signal)
        range_val = np.ptp(signal)
        features = {
        'Mean': mean,
        'Median': median,
        'Standard Deviation': std_dev,
        'Variance': variance,
        'Skewness': skewness,
        'Kurtosis': kurtosis,
        'Range': range_val
}
        return features
```


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Statistical features usage

```
# Extract statistical features
stat_features = extract_statistical_features(ecg_signal)
print("Statistical features:", stat_features)
```

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Texture Features

- Contrast: $\sum_{i,j}(i-j)^2 P(i,j)$, where $P(i,j)$ is the co-occurrence matrix.
- Correlation: $\sum_{i,j} \frac{(i-\mu_i)(j-\mu_j)P(i,j)}{\sigma_i \sigma_j}$, where μ_i and μ_j are the means and σ_i and σ_j are the standard deviations of i and j.
- Energy: $\sum_{i,j} P(i,j)^2$, where $P(i,j)$ is the co-occurrence matrix.
- Homogeneity: $\sum_{i,j} \frac{P(i,j)}{1+|i-j|}$, where $P(i,j)$ is the co-occurrence matrix.
- Entropy: $-\sum_{i,j} P(i,j) \log P(i,j)$, where $P(i,j)$ is the co-occurrence matrix.

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Related library

```
# Related library
from skimage.feature import greycomatrix, greycoprops
from skimage.measure import shannon_entropy
```

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Texture features examples

```
# Texture Features
def extract_texture_features(signal):
# Convert signal to 2D array for texture analysis
        signal_2d = np.reshape(signal, (int(np.sqrt(len(signal))), -1))

# Compute grey-level co-occurrence matrix (GLCM)
        glcm = greycomatrix(signal_2d, distances=[1], angles=[0], levels
             =256, symmetric=True, normed=True)
# Extract texture features
        contrast = greycoprops(glcm, 'contrast')[0,0]
        dissimilarity = greycoprops(glcm, 'dissimilarity')[0,0]
        homogeneity = greycoprops(glcm, 'homogeneity')[0, 0]
        energy = greycoprops(glcm, 'energy')[0, 0]
        correlation = greycoprops(glcm, 'correlation')[0, 0]
        asm = greycoprops(glcm, 'ASM')[0, 0]
        entropy = shannon_entropy(signal_2d)

        features = {
        'Contrast': contrast,
        'Dissimilarity': dissimilarity,
        'Homogeneity': homogeneity,
        'Energy': energy,
        'Correlation': correlation,
        'ASM': asm,
        'Entropy': entropy
}
        return features
```


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Texture features usage

```
texture_features = extract_texture_features(ecg_signal)
print("Texture features:", texture_features)
```

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Entropy Features

- Shannon Entropy: $H(X) = -\sum_i p(x_i) \log p(x_i)$, where $p(x_i)$ is the probability of occurrence of x_i.
- Renyi Entropy: $H_\alpha(X) = \frac{1}{1-\alpha} \log \sum_i p(x_i)^\alpha$, where α is a parameter and $p(x_i)$ is the probability of occurrence of x_i.
- Tsallis Entropy: $H_q(X) = \frac{1}{q-1} (1 - \sum_i p(x_i)^q)$, where q is a parameter and $p(x_i)$ is the probability of occurrence of x_i.
- Differential Entropy: $h(X) = -\int p(x) \log p(x)\, dx$, where $p(x)$ is the probability density function.
- Cross Entropy: $H(p, q) = -\sum_i p(x_i) \log q(x_i)$, where $p(x_i)$ and $q(x_i)$ are the probability distributions.

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Entropy features examples

```python

# Entropy Features
def extract_entropy_features(signal):
        sample_entropy = ant.sample_entropy(signal)
        spectral_entropy = ant.spectral_entropy(signal, sf=fs, method='welch')
        perm_entropy = ant.perm_entropy(signal, normalize=True)
        svd_entropy = ant.svd_entropy(signal, order=3, delay=1)
        app_entropy = ant.app_entropy(signal)
        lziv_complexity = ant.lziv_complexity(signal)

        features = {
                'Sample Entropy': sample_entropy,
                'Spectral Entropy': spectral_entropy,
                'Permutation Entropy': perm_entropy,
                'SVD Entropy': svd_entropy,
                'Approximate Entropy': app_entropy,
                'LZiv Complexity': lziv_complexity
                }

        return features
```


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Entropy features usage

```
entropy_features = extract_entropy_features(ecg_signal)
print("Entropy features:", entropy_features)
```

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Image Features

- Edges (using Sobel, Canny, etc.)
- Corners (using Harris, Shi-Tomasi, etc.)
- Blobs (using Laplacian of Gaussian, Difference of Gaussian, etc.)
- Maximum intensity
- Minimum intensity
- Mean intensity
- Standard deviation of intensity

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Image features examples

```
# Image Features
def extract_image_features(signal):
# Convert signal to 2D array for image analysis
        signal_2d = np.reshape(signal, (int(np.sqrt(len(signal))), -1))

# Extract image features
        edges = np.sum(np.abs(np.diff(signal_2d, axis=0))) + np.sum(np.
             abs(np.diff(signal_2d, axis=1)))
        corners = np.sum(signal_2d > np.mean(signal_2d) + 2 * np.std(
             signal_2d))
        blobs = np.sum(signal_2d > np.mean(signal_2d) + np.std(signal_2d
             ))
        mean_intensity = np.mean(signal_2d)
        std_intensity = np.std(signal_2d)
        max_intensity = np.max(signal_2d)
        min_intensity = np.min(signal_2d)

        features = {
                'Edges': edges,
                'Corners': corners,
                'Blobs': blobs,
                'Mean Intensity': mean_intensity,
                'Standard Deviation Intensity': std_intensity,
                'Max Intensity': max_intensity,
                'Min Intensity': min_intensity
        }

        return features
```


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Image features usage

```
image_features = extract_image_features(ecg_signal)
print("Image features:", image_features)
```

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Frequency Domain Features

- Fourier Transform: $X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft}\, dt$, where $x(t)$ is the time-domain signal and $X(f)$ is the frequency-domain representation.
- Wavelet Transform: $W_\psi(a, b) = \int_{-\infty}^{\infty} x(t) \psi^* \left(\frac{t-b}{a}\right)\, dt$, where $x(t)$ is the signal, ψ is the wavelet function, a is the scale, and b is the translation.
- Power Spectral Density (PSD): Describes how the power of a signal is distributed with frequency.
- Cepstrum: $C(x) = \mathcal{F}^{-1}(\log |\mathcal{F}(x)|)$, where $\mathcal{F}$ denotes the Fourier transform.
- Mel-Frequency Cepstral Coefficients (MFCCs): Coefficients that represent the short-term power spectrum of a sound.

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Frequency domain features examples

```
# Frequency Domain Features
def extract_frequency_domain_features(signal, fs):
        freqs, psd = signal.welch(signal, fs)
        dominant_freq = freqs[np.argmax(psd)]
        total_power = np.sum(psd)
        band_power = np.sum(psd[(freqs >= 0.5) & (freqs <= 40)])
        mean_freq = np.mean(freqs)
        median_freq = np.median(freqs)
        peak_freq = freqs[np.argmax(psd)]
        freq_variance = np.var(freqs)

    features = {
        'Dominant Frequency': dominant_freq,
        'Total Power': total_power,
        'Band Power (0.5-40 Hz)': band_power,
        'Mean Frequency': mean_freq,
        'Median Frequency': median_freq,
        'Peak Frequency': peak_freq,
        'Frequency Variance': freq_variance
}
        return features
```


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Frequency domain features usage

```
freq_features = extract_frequency_domain_features(ecg_signal, fs)
print("Frequency domain features:", freq_features)
```

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Time Domain Features

- Temporal Moments: Statistical moments calculated over time.
- Zero Crossing Rate: $ZCR = \frac{1}{N-1} \sum_{i=1}^{N-1} \mathbb{1}_{\{x_i x_{i+1} < 0\}}$, where N is the number of samples and $\mathbb{1}$ is the indicator function.
- Autocorrelation: $R(\tau) = \sum_t x(t)x(t+\tau)$, where $x(t)$ is the signal and τ is the lag.
- Root Mean Square (RMS): $RMS = \sqrt{\frac{1}{N} \sum_{i=1}^{N} x_i^2}$, where N is the number of samples and x_i is the i-th sample.
- Signal Energy: $E = \sum_{i=1}^{N} x_i^2$, where N is the number of samples and x_i is the i-th sample.

Time Domain features examples

```
# Time Domain Features
def extract_time_domain_features(signal):
        rms = np.sqrt(np.mean(signal**2))
        zero_crossings = ((signal[:-1] * signal[1:]) < 0).sum()
        autocorrelation = np.correlate(signal, signal, mode='full')[len(
             signal)-1]
        mean_abs_dev = np.mean(np.abs(signal - np.mean(signal)))
        max_val = np.max(signal)
        min_val = np.min(signal)
        signal_energy = np.sum(signal**2)

        features = {
        'RMS': rms,
        'Zero Crossings': zero_crossings,
        'Autocorrelation': autocorrelation,
        'Mean Absolute Deviation': mean_abs_dev,
        'Max Value': max_val,
        'Min Value': min_val,
        'Signal Energy': signal_energy
}
        return features
```


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Time domain features usage

```
time_features = extract_time_domain_features(ecg_signal)
print("Time domain features:", time_features)
```

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How Features Reduce Dimensionality

- Feature extraction transforms data into a lower-dimensional space.
- Reduces the number of input variables, making models simpler and faster.
- Helps in removing redundant and irrelevant data.
- Techniques include Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA), and t-Distributed Stochastic Neighbor Embedding (t-SNE).

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Feature Concatenation

- Combines information from multiple sources.
- Enhances the model's ability to learn complex patterns.
- Useful in scenarios with heterogeneous data types.
- Consider two feature vectors: **X1** and **X2**.
- Concatenation results in a new feature vector: $\mathbf{X} = [\mathbf{X1}, \mathbf{X2}]$.

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Feature concatenation example and usage

```
# Combine all features into a single DataFrame
def combine_features(stat_features, time_features, freq_feature,
    entropy_features):
combined_features_list = []
for i in range(len(stat_features)):
combined_features = {**stat_features[i], **time_features[i], **
    freq_features[i], **entropy_features[i], }
combined_features_list.append(combined_features)
return pd.DataFrame(combined_features_list)

combined_features_df = combine_features(stat_features, time_features,
    freq_features, entropy_features)
print("Combined Features DataFrame:")
print(combined_features_df)
```

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Feature normalization example and usage

```
from sklearn.preprocessing import StandardScaler
# Create DataFrame
df = pd.DataFrame(combined_features_df)

# Initialize the StandardScaler
scaler = StandardScaler()

# Apply the scaler to normalize the data
normalized_data = scaler.fit_transform(df)

# Convert the result back into a DataFrame
normalized_df = pd.DataFrame(normalized_data, columns=
    combined_features_df.columns)
print(normalized_df)
```


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Feature Reduction: Importance

- **Feature reduction** refers to reducing the number of input variables in a dataset.
- Helps to improve model performance by removing irrelevant or redundant features.
- Reduces computation time and storage space.
- Can prevent overfitting in machine learning models.
- Makes it easier to visualize high-dimensional data.

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Techniques for Feature Reduction

- **Principal Component Analysis (PCA)**: Linear transformation that reduces dimensionality by projecting data onto the directions of maximum variance.
- **Linear Discriminant Analysis (LDA)**: Maximizes the separation between classes while reducing dimensions.
- **t-Distributed Stochastic Neighbor Embedding (t-SNE)**: Non-linear technique used for the visualization of high-dimensional datasets.
- **Autoencoders**: Neural network-based technique to compress and reconstruct the data, often used for non-linear feature reduction.
- **Feature Selection Methods**: Such as Recursive Feature Elimination (RFE), SelectKBest, and others.

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Principal Component Analysis (PCA) Overview

- PCA is a statistical method for reducing the number of dimensions (features) in a dataset.
- It transforms the original features into a new set of uncorrelated features (principal components).
- The components are ordered by the amount of variance they capture from the data.
- PCA seeks to find the directions of maximum variance in the data and project the data along those directions.

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PCA: Algorithm and Mathematical Explanation

- Step 1: Standardize the Data Standardize the dataset to have a mean of 0 and a variance of 1 for each feature.

$$X_{std} = \frac{X - \mu}{\sigma}$$

  where X is the dataset, μ_x is the mean, and σ is the standard deviation of each feature.

- Step 2: Compute the Covariance Matrix between two random variables X and Y. The covariance matrix shows how the features vary together.

$$C(X, Y) = \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu_x)(y_i - \mu_y)$$

  where:

  - N is the number of observations.
  - X_i and Y_i are the individual observations of the random variables X and Y.
  - μ_x and μ_y is the mean of the observations of X and (Y)

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PCA: Algorithm and Mathematical Explanation

- Step 3: Compute the Eigenvectors and Eigenvalues Eigenvectors (principal components) and their corresponding eigenvalues are computed from the covariance matrix.

$$Cv = \lambda v$$

  where v is the eigenvector and λ is the eigenvalue.
- Step 4: Sort Eigenvalues Sort the eigenvalues in descending order. The eigenvectors corresponding to the largest eigenvalues are the most important principal components.
- Step 5: Choose Principal Components Select the top k eigenvectors corresponding to the largest eigenvalues, forming a new feature space.

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Physical Interpretation of PCA

- PCA can be interpreted as finding the "directions" in the data that capture the most variance.
- The first principal component represents the direction in the feature space where the data varies the most.
- Each subsequent component is orthogonal to the previous one, capturing the remaining variance.
- Geometrically, PCA projects the data points onto a lower-dimensional plane or line while retaining the most important relationships in the data.

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Advantages and Disadvantages of PCA

Advantages:

- Reduces dimensionality, improving computational efficiency.
- Removes multicollinearity by creating uncorrelated features.
- Retains most of the information (variance) with fewer components.
- Helps in visualizing high-dimensional data (e.g., 2D or 3D projection).

Disadvantages:

- PCA is sensitive to scaling of the data.
- The new features (principal components) may not be easily interpretable.
- Assumes linearity in the data, not suitable for non-linear relationships.
- Requires choosing the number of components to retain, which may not always be straightforward.

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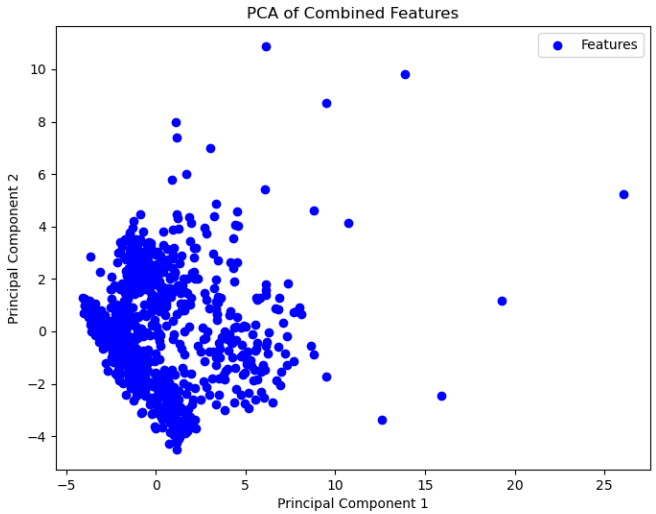
PCA example and usage

```python
# Import necessary libraries
from sklearn.decomposition import PCA
import seaborn as sns

# Apply PCA
pca = PCA(n_components=2)
pca_result = pca.fit_transform(normalized_df)
```



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Kruskal Wallis example and usage

```
# Generate synthetic labels for two classes
# Perform Kruskal-Wallis test
kruskal_results = {}
for feature in normalized_df.columns:
        class_0 = normalized_df[feature][:400]
        class_1 = normalized_df[feature][401:]
        kruskal_results[feature] = stats.kruskal(class_0, class_1).
             pvalue
# Export Kruskal-Wallis test results to Excel
kruskal_results_df = pd.DataFrame(list(kruskal_results.items()), columns
     =['Feature', 'P-value'])
kruskal_results_df.to_excel('kruskal_wallis_results.xlsx', index=False)
print("Kruskal-Wallis test results have been exported to '
     kruskal_wallis_results.xlsx'.")

# Plot box plots for each feature
plt.figure(figsize=(12, 8))
features_to_plot = ['LZiv Complexity', 'Standard Deviation', 'Min Value'
     , 'Sample Entropy']
normalized_df['Class'] = Y
melted_df = pd.melt(normalized_df, id_vars='Class', var_name='Feature',
     value_name='Value')
melted_df = melted_df[melted_df['Feature'].isin(features_to_plot)]
sns.boxplot(x='Feature', y='Value', hue='Class', data=melted_df)
plt.xticks(rotation=90)
plt.title('Box Plot of Two Classes')
plt.show()
```


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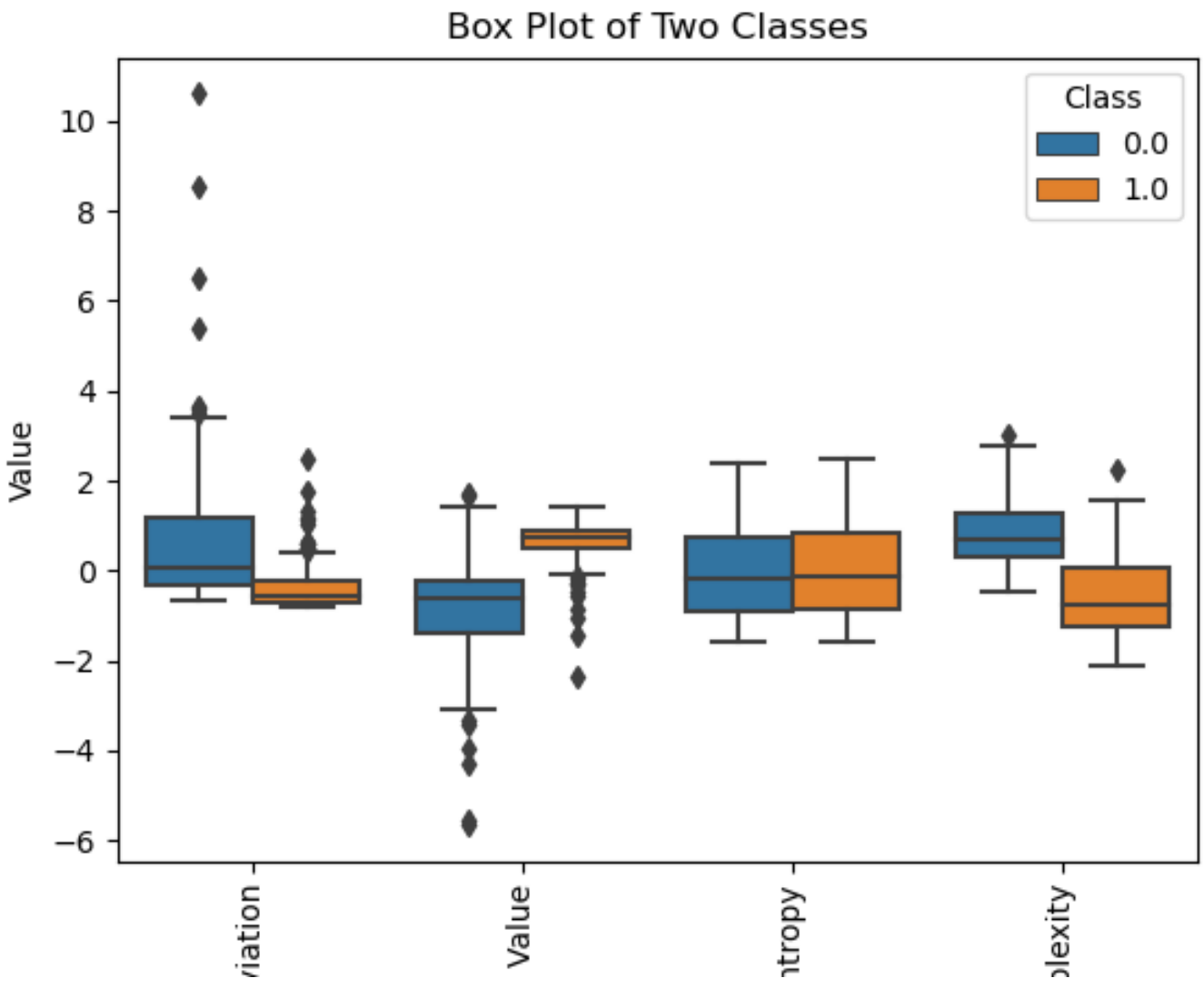
Kruskal Wallis example and usage



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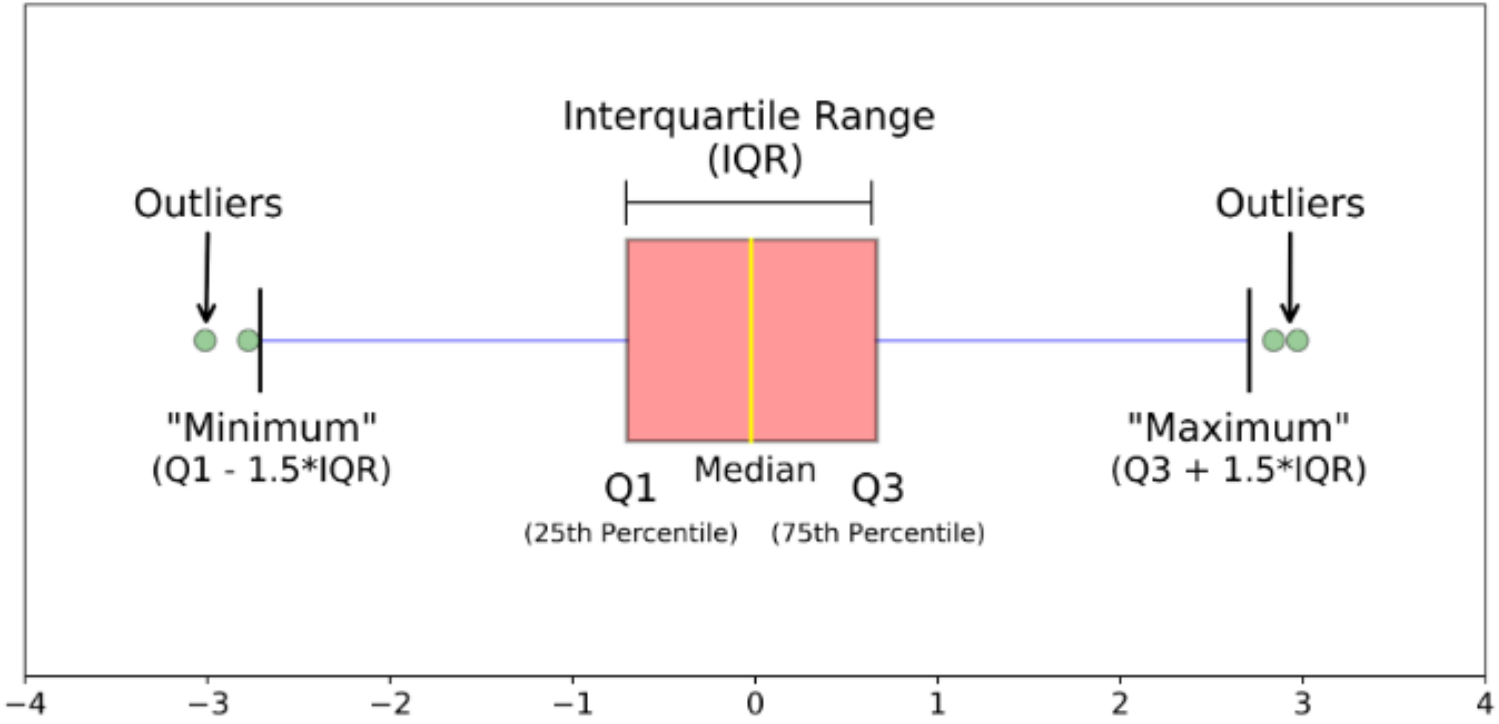
Box plot explained



- **Median (Q2/50th percentile)**: The middle value of the data set.
- **First Quartile (Q1/25th percentile)**: The middle number between the smallest number (not the "minimum") and the median of the data set.
- **Third Quartile (Q3/75th percentile)**: The middle value between the median and the highest value (not the "maximum") of the dataset.
- **Interquartile Range (IQR)**: 25th to the 75th percentile.
- **Whiskers** (shown in blue)
- **Outliers** (shown as green circles)
- "**Minimum**": Q1 - 1.5*IQR
- "**Maximum**": Q3 + 1.5*IQR

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Introduction to SVM

- Support Vector Machines (SVM) are supervised learning models used for classification and regression tasks.
- SVMs aim to find the optimal hyperplane that maximizes the margin between different classes.
- They are effective in high-dimensional spaces and are versatile with different kernel functions.

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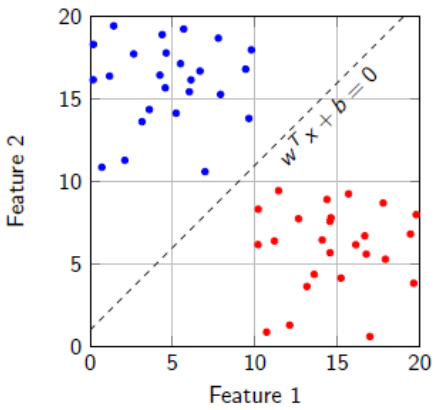
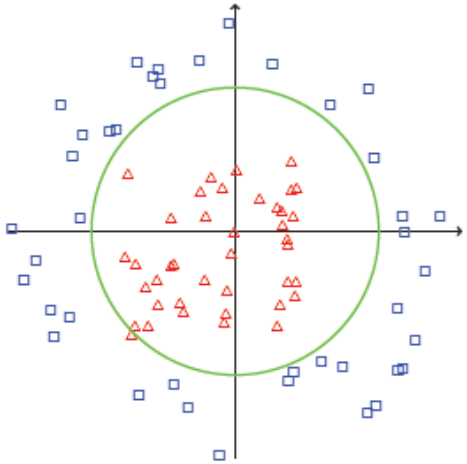
Linear and Nonlinear Problems

- **Linear Problems**: Problems where data points can be separated by a straight line (2D) or a hyperplane (higher dimensions).
- **Nonlinear Problems**: Data points cannot be separated by a single straight line or hyperplane.
- **Kernel Trick**: SVM maps nonlinear problems into higher dimensions to make them linearly separable using kernels (e.g., polynomial, radial basis function (RBF)).



(a) Linear



(b) Nonlinear

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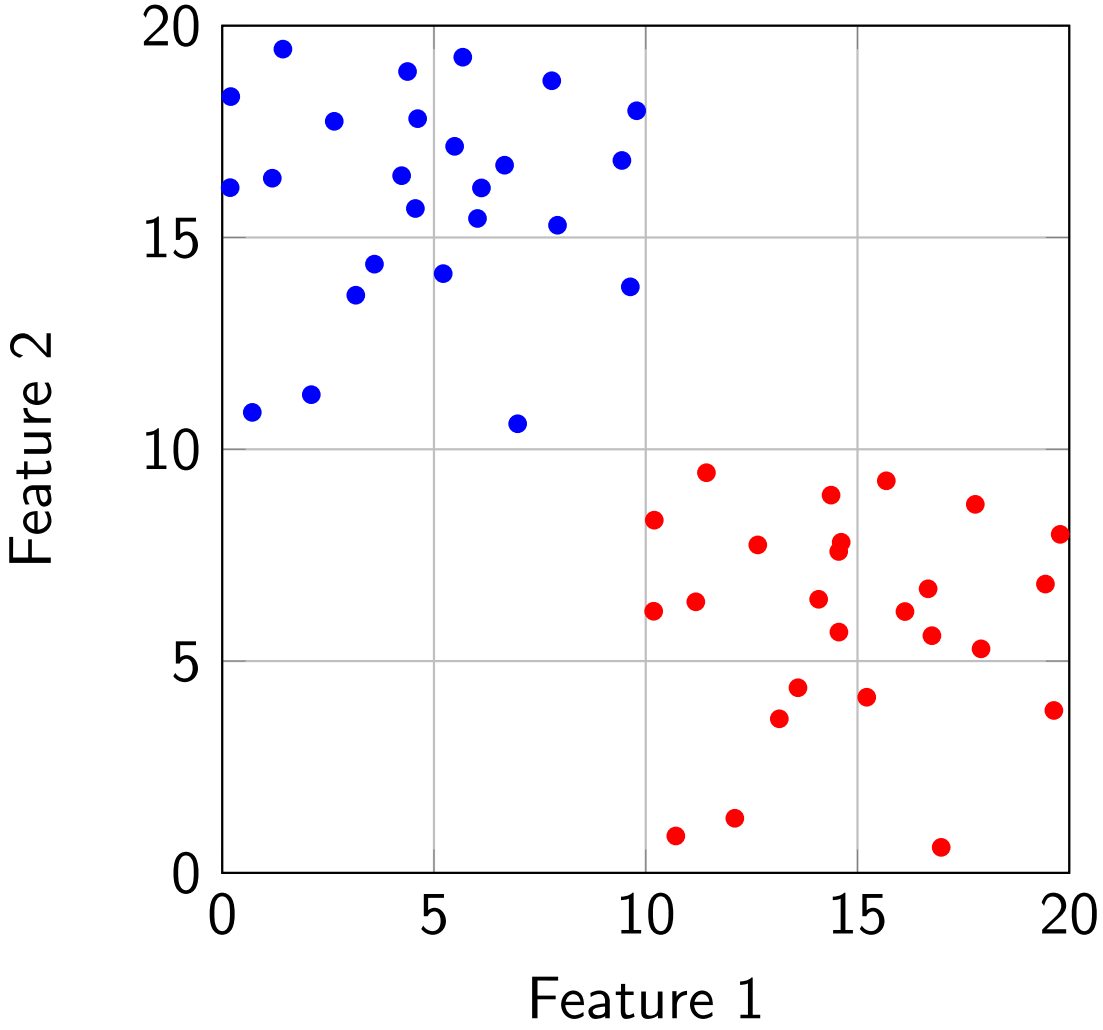
Perceptron revisited: Linear separators



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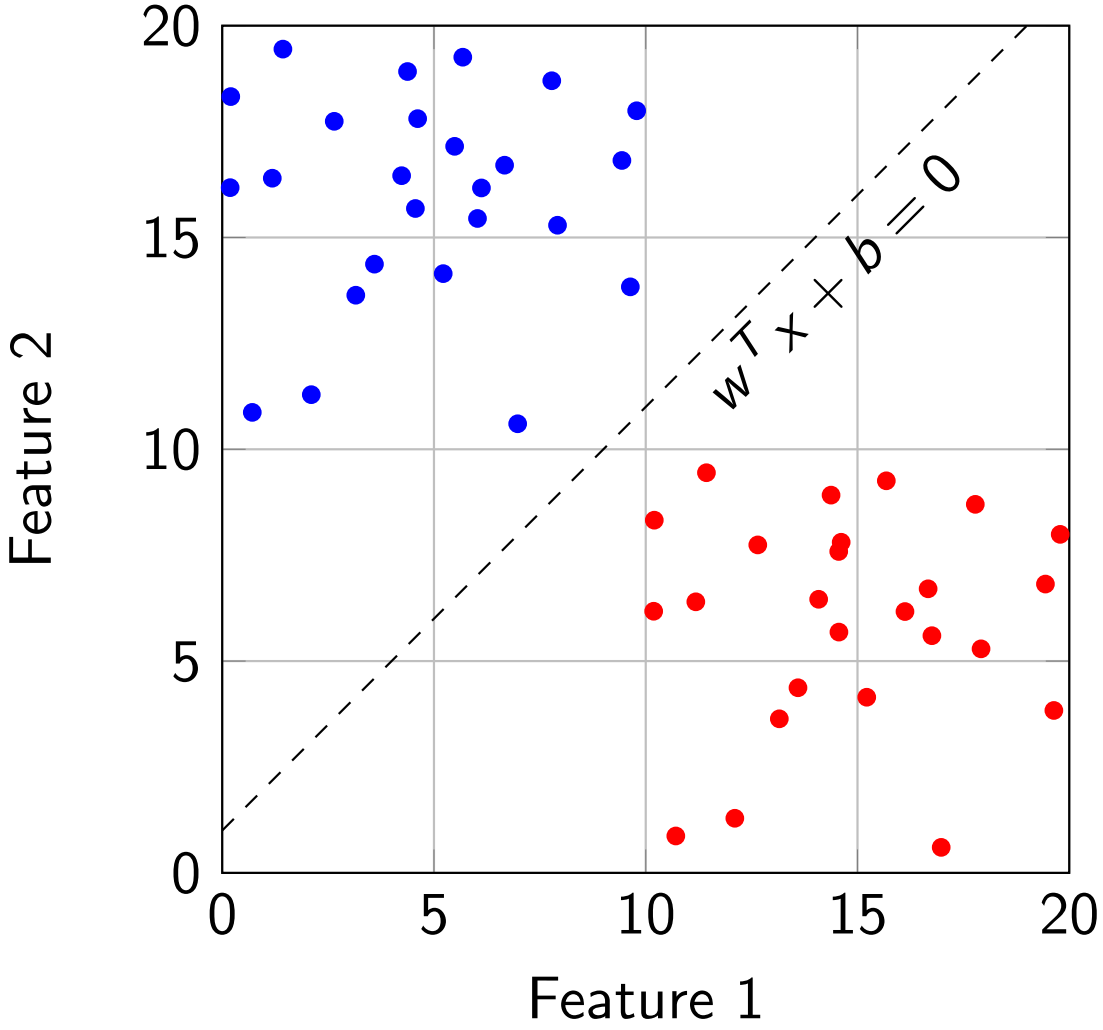
Perceptron revisited: Linear separators



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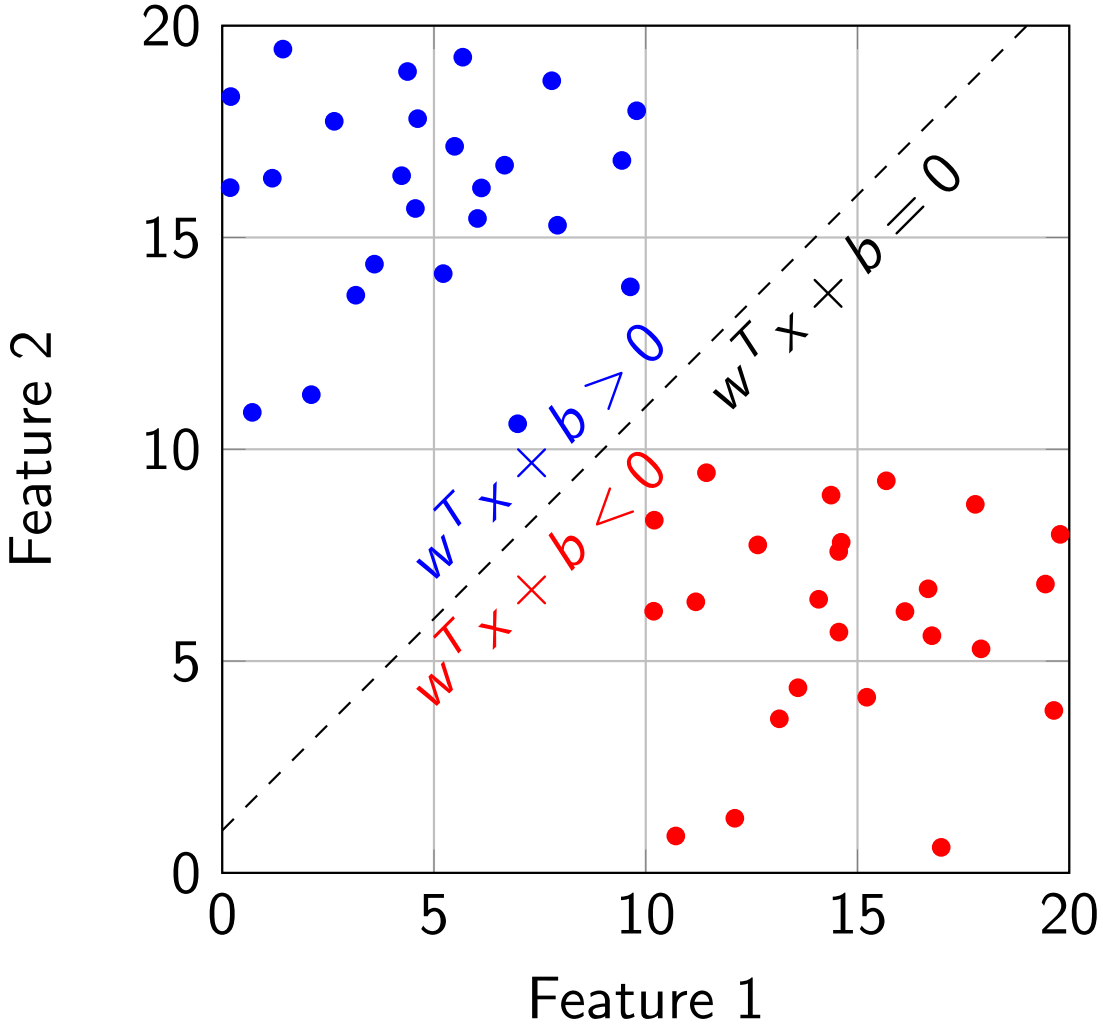
Perceptron revisited: Linear separators



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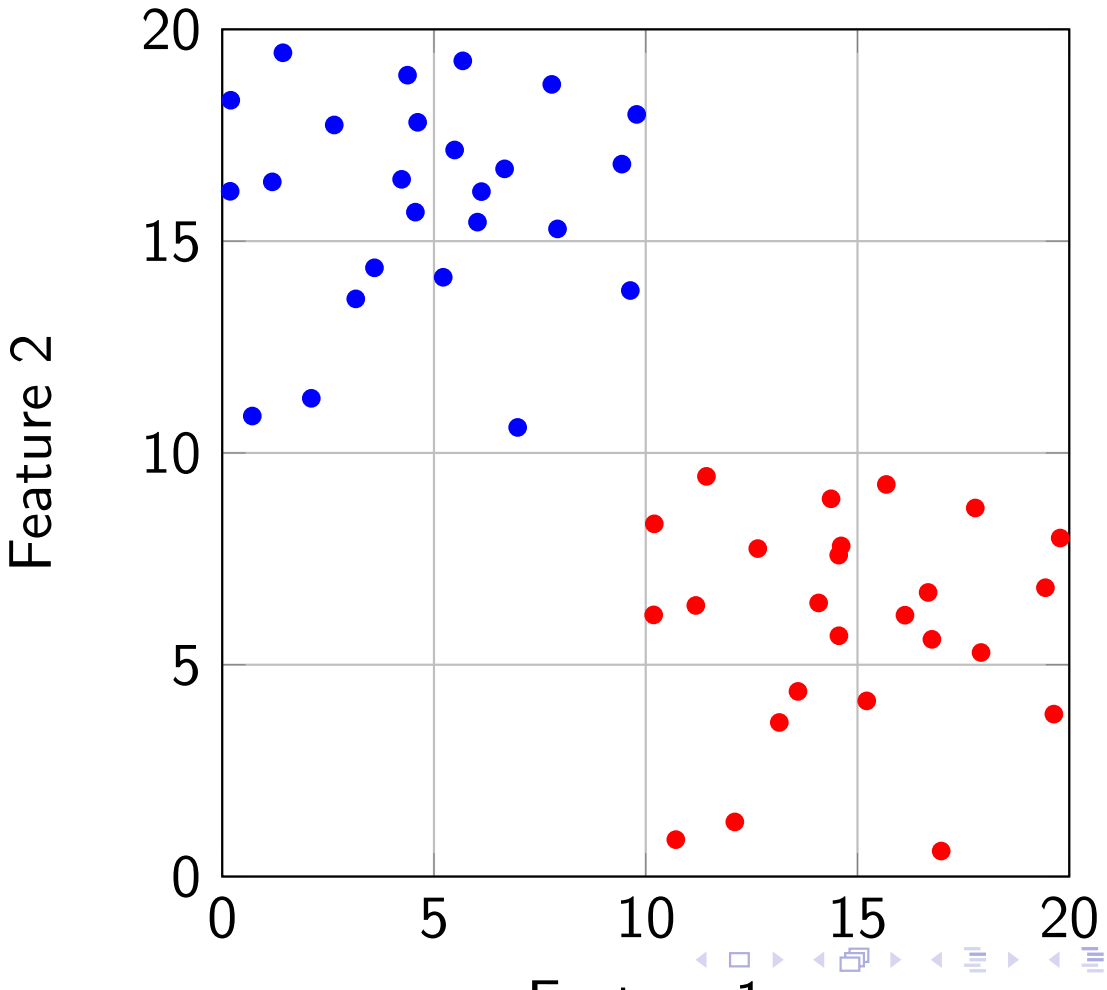
Multiple Decision Boundaries

- Multiple decision boundaries can exist when exploring different hyperplanes.
- Each boundary may correspond to a different separation strategy.



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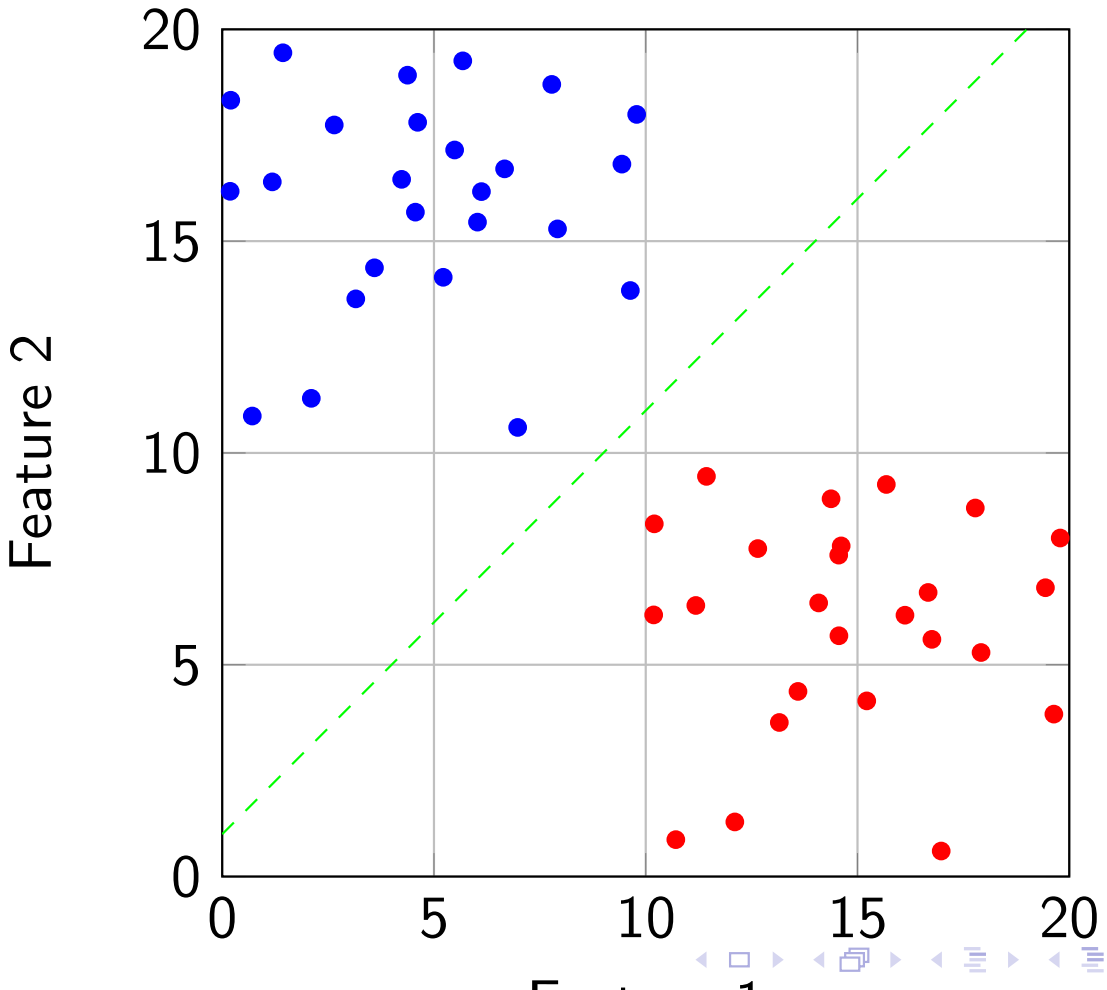

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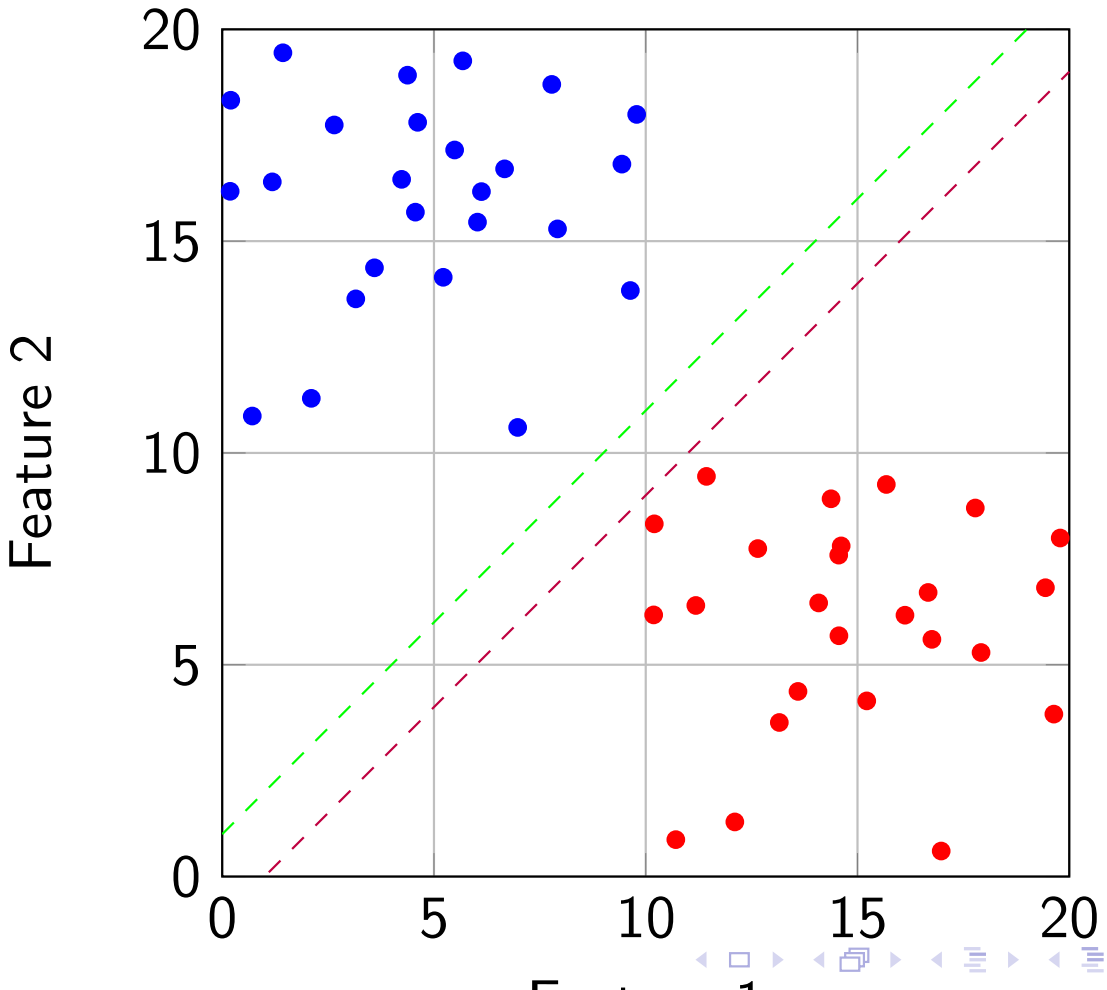

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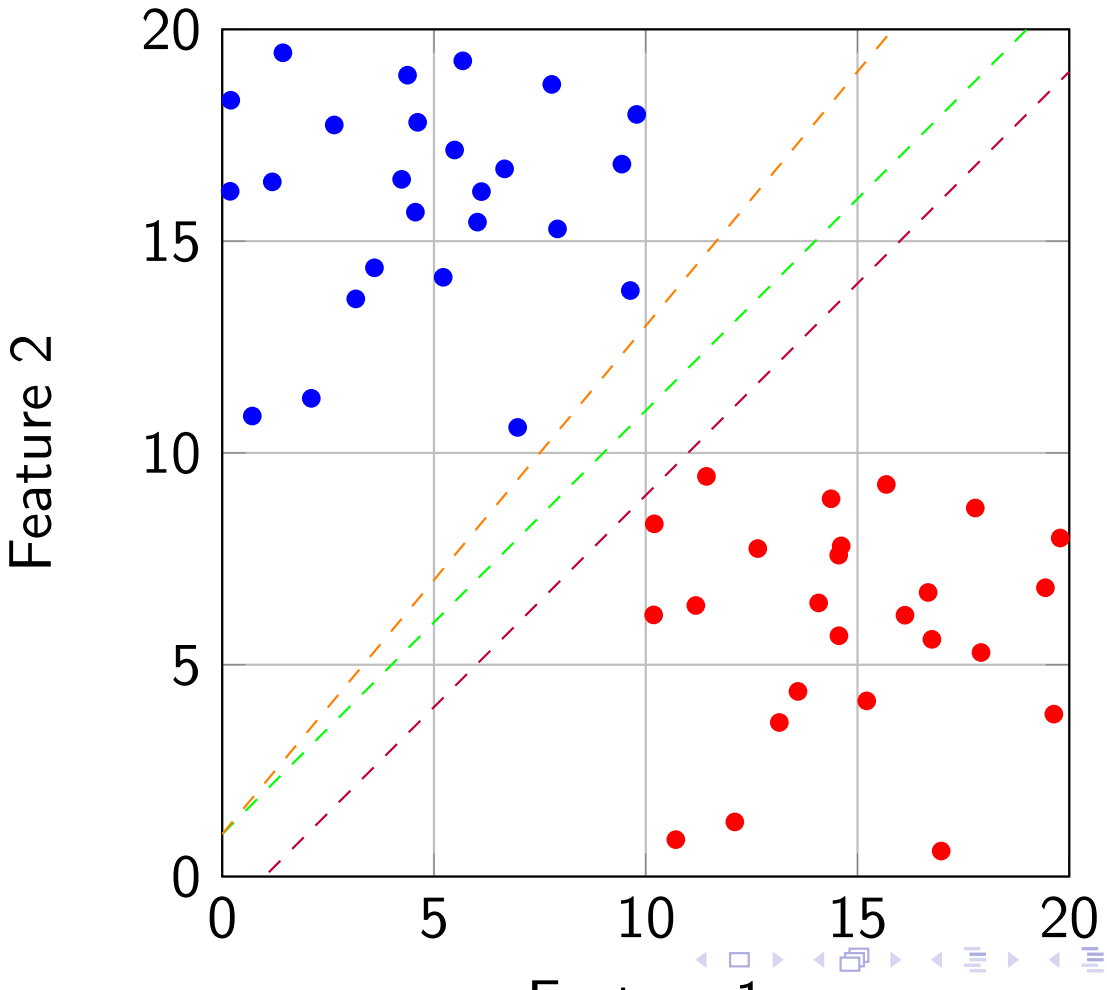

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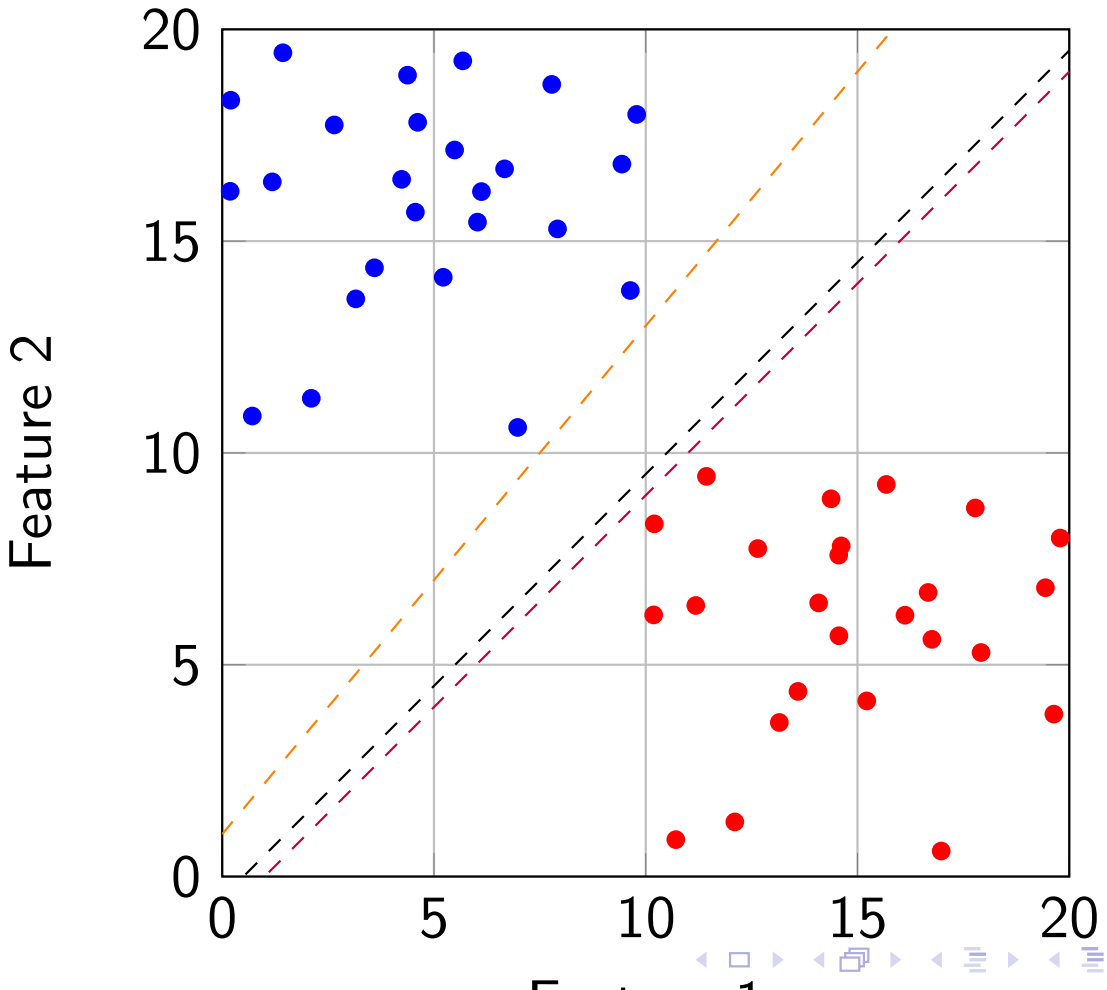

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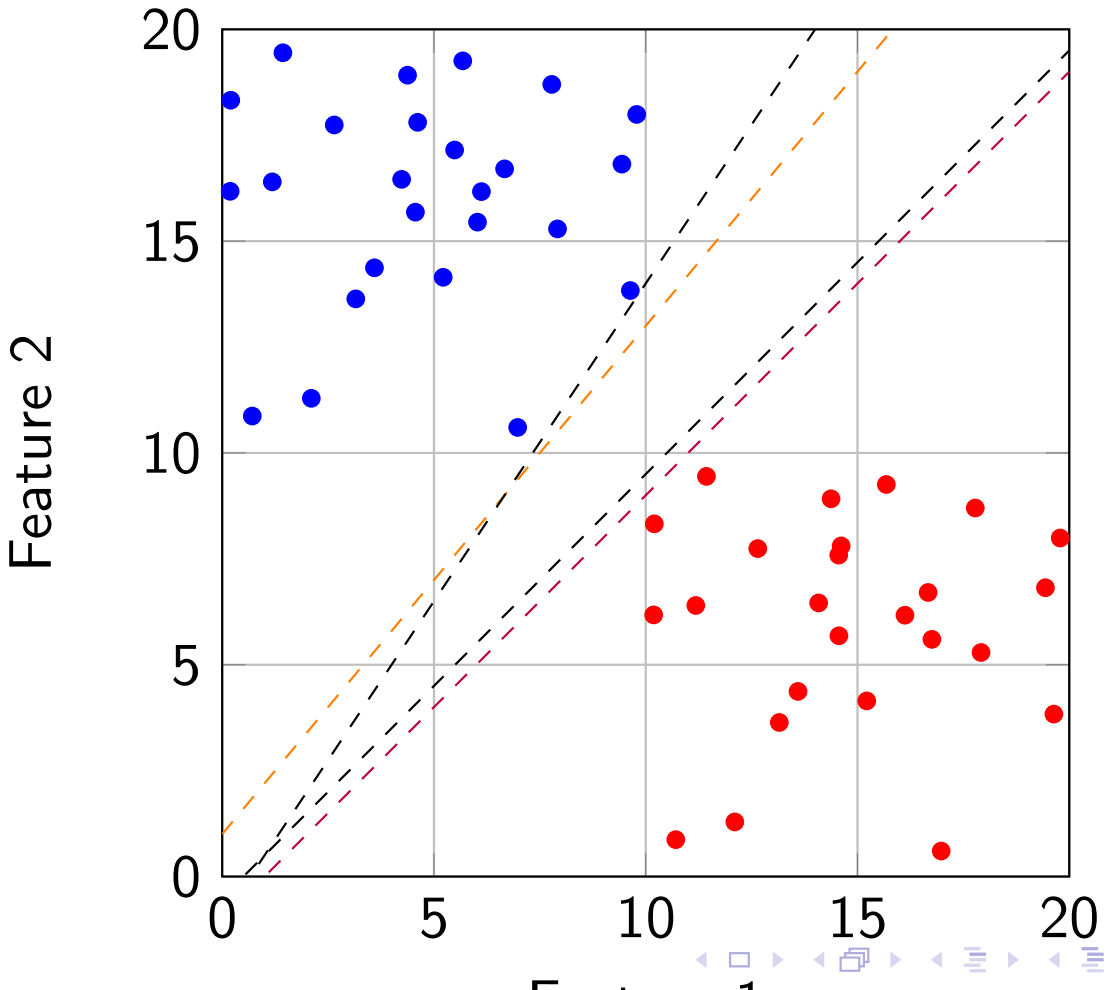

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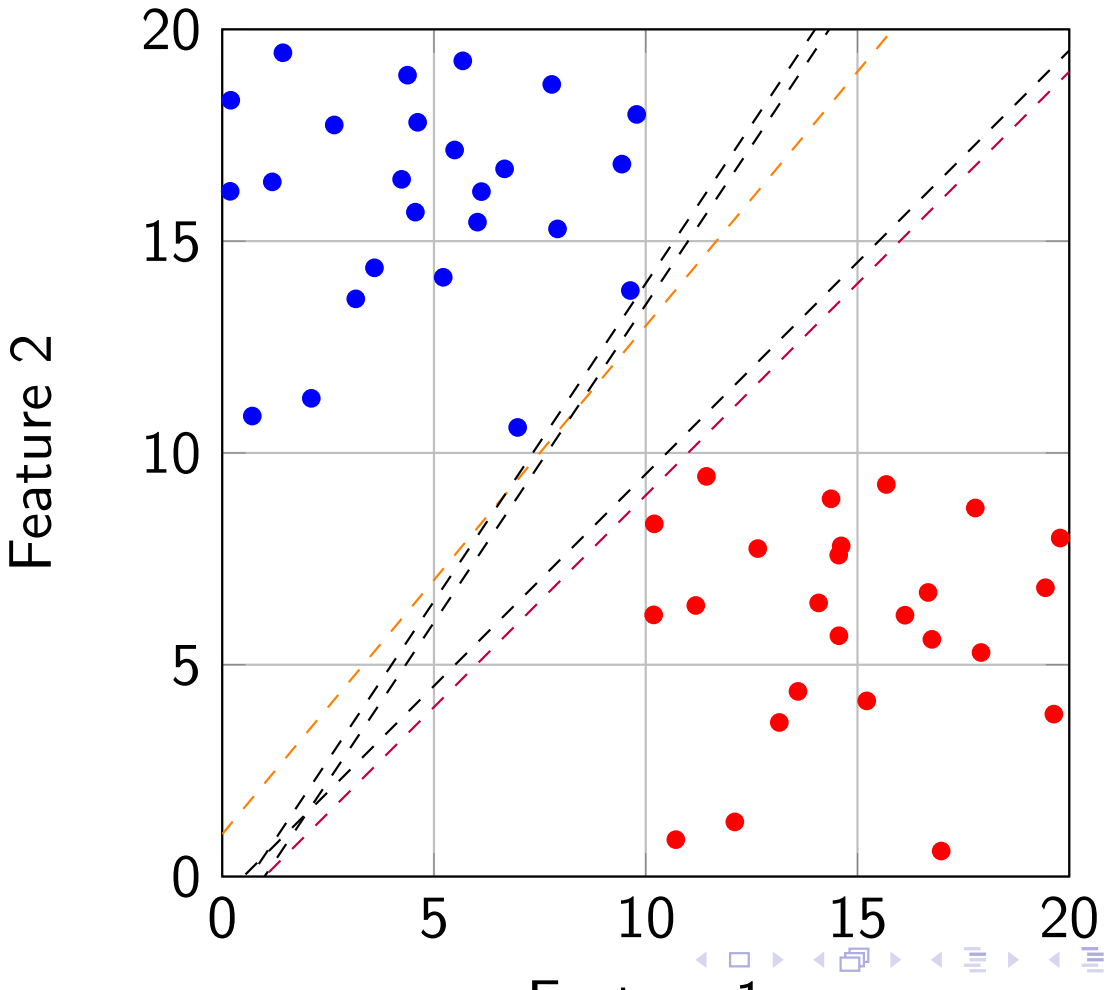

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Hyperplane and Classification Margin

- **Hyperplane**: A decision boundary that separates data points of different classes.
- **Classification Margin**: The distance between the hyperplane and the closest data points (support vectors).
- **Maximum Margin Classification**: SVM aims to maximize this margin for better generalization.
- A larger margin is preferred as it indicates better generalization to unseen data.
- The goal is to maximize the margin while correctly classifying the training data.

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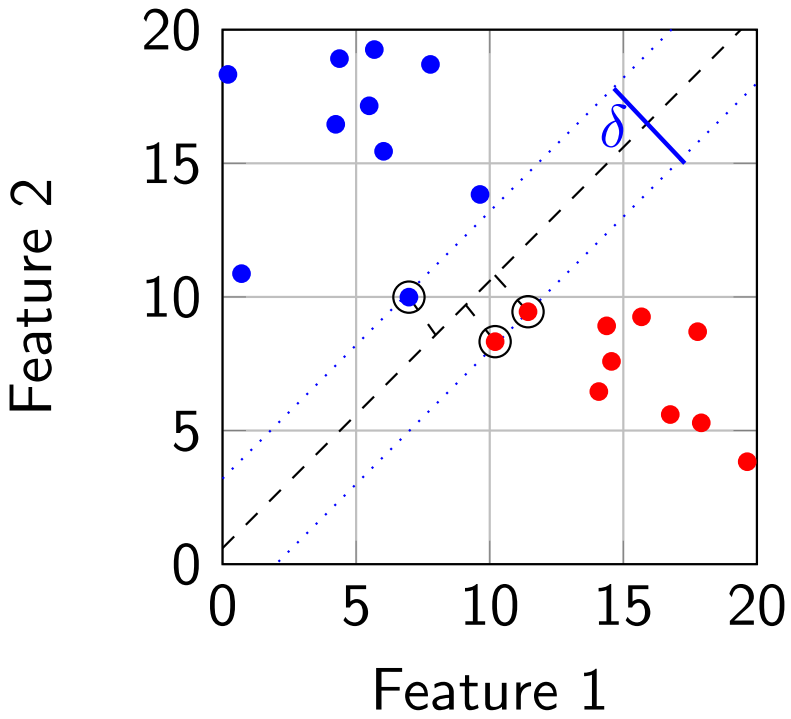
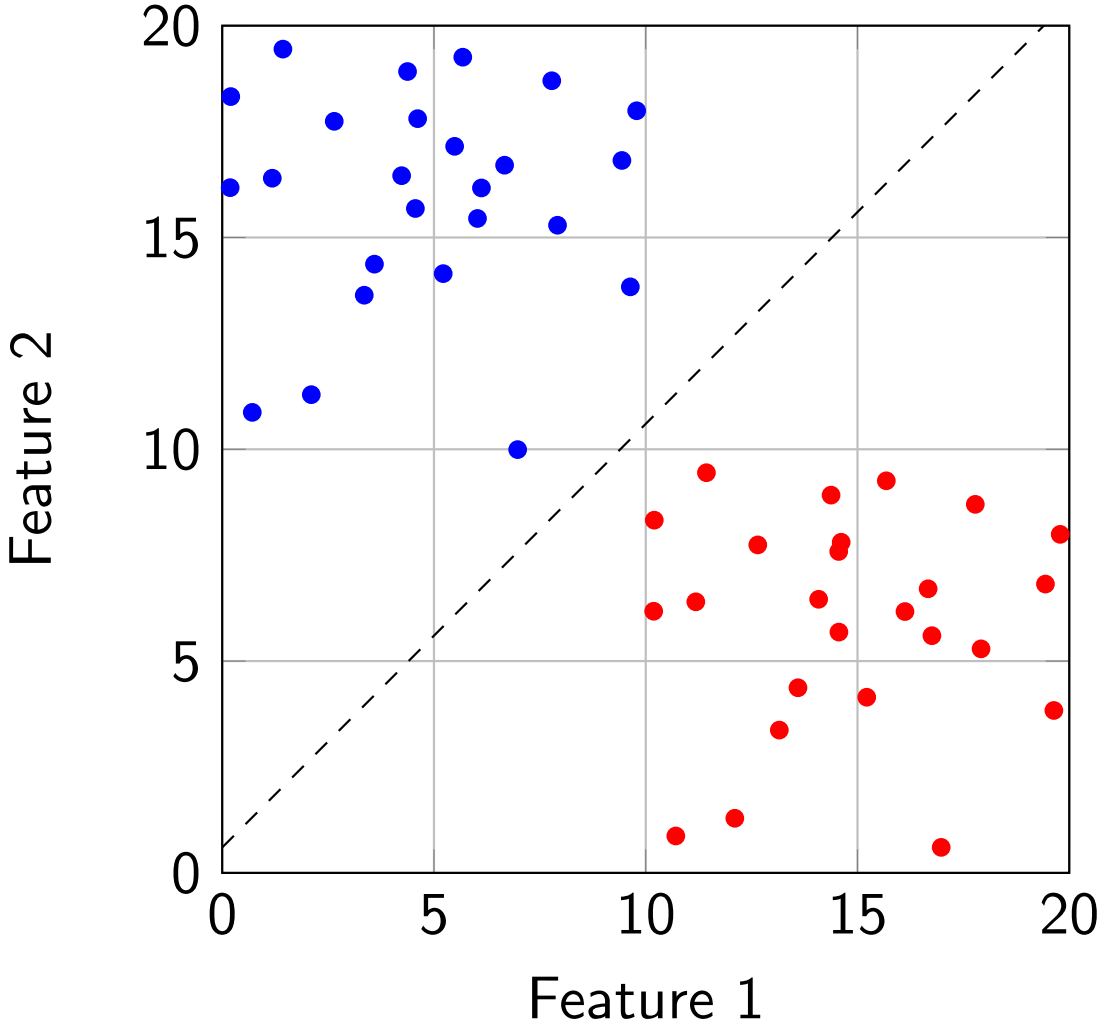
Visualizing the Margin



- **Hyperplane (Decision Boundary):** The black dashed line represents the decision boundary separating the two groups.
- **Margin (δ):** The region between the blue dotted lines. It is the distance between the hyperplane and the closest data points (support vectors).
- **Support Vectors:** Points that lie on the margin lines. For example, the points circled in black at coordinates.

Visualizing the Margin



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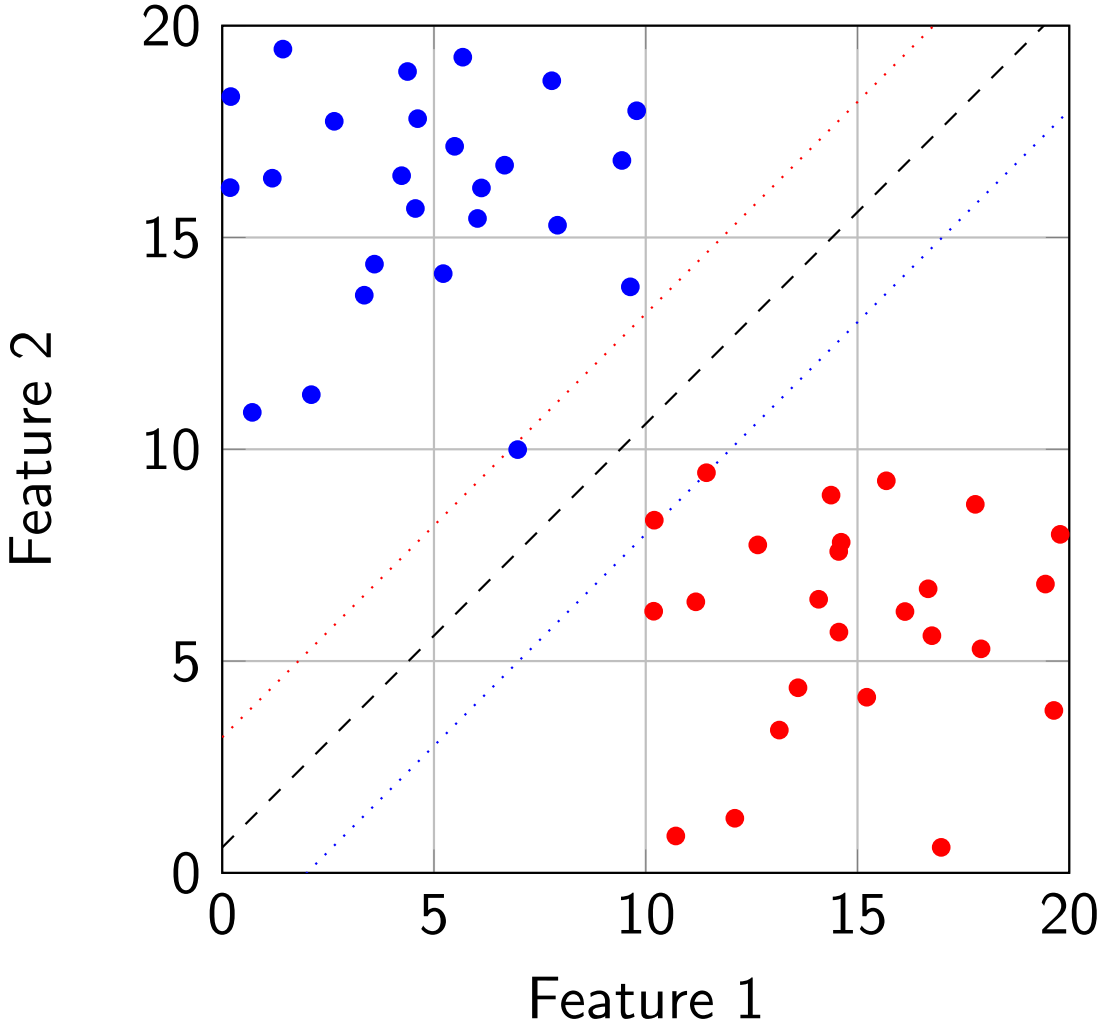
Visualizing the Margin



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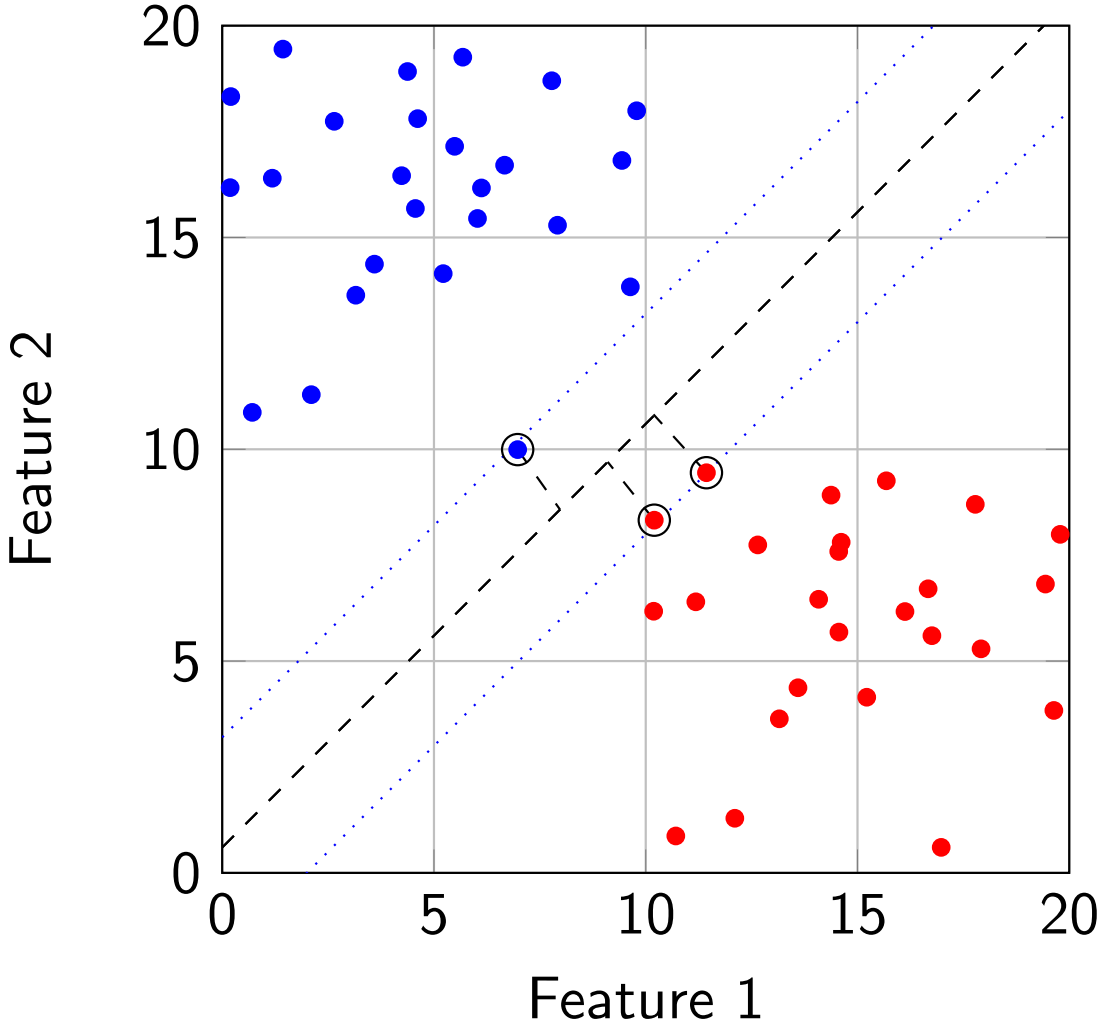

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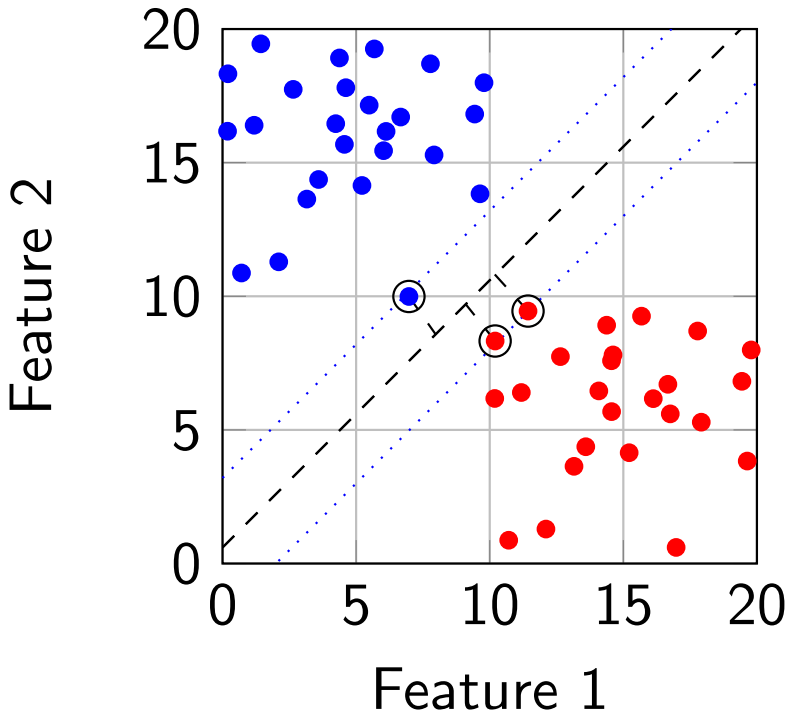
Soft Margin Classification

- **Hard Margin**: Assumes data is perfectly separable. Real-world data often contains noise or overlap between classes.



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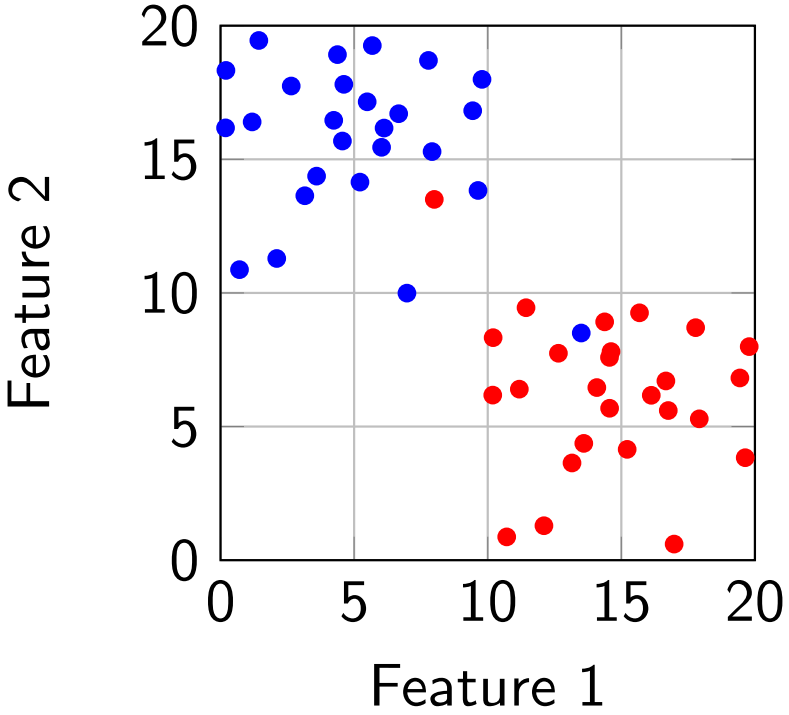
Soft Margin Classification



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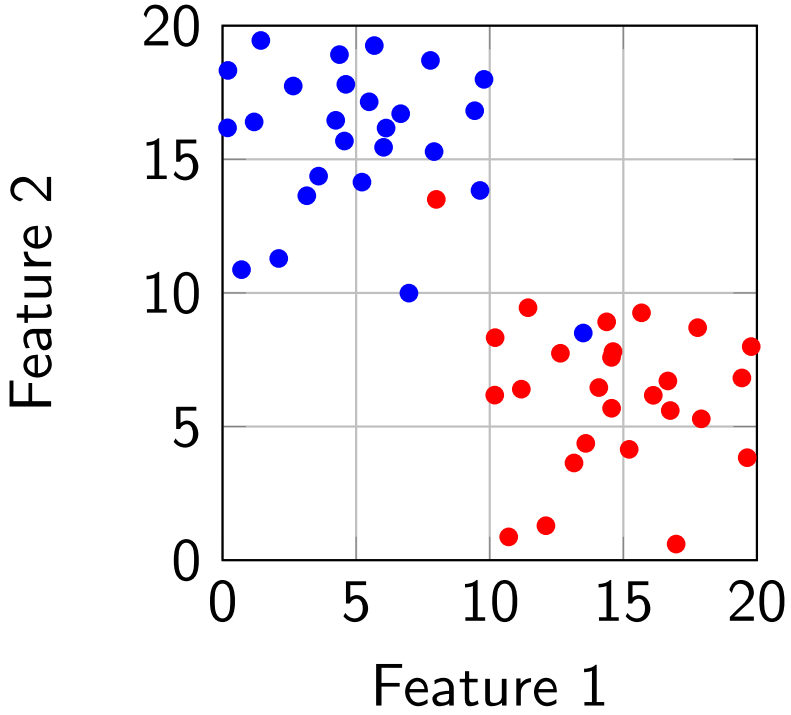
Soft Margin Classification



- What if the training set is not linearly separable?

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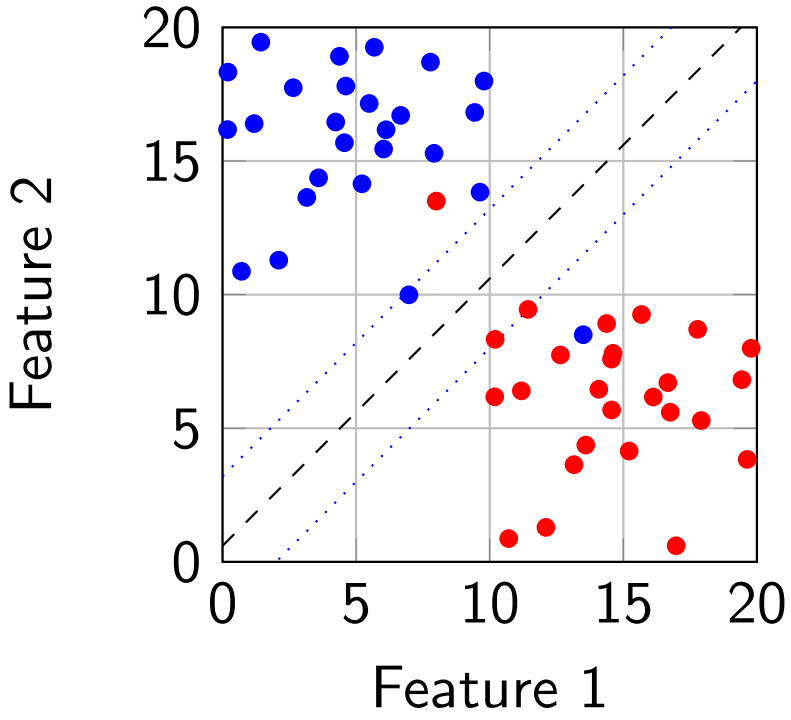
Soft Margin Classification

- **Soft Margin**: Allows for some misclassification to handle noisy or overlapping data.
- **Slack variables (ξ_i)** can be added to allow misclassification of difficult or noisy examples, resulting margin called soft.



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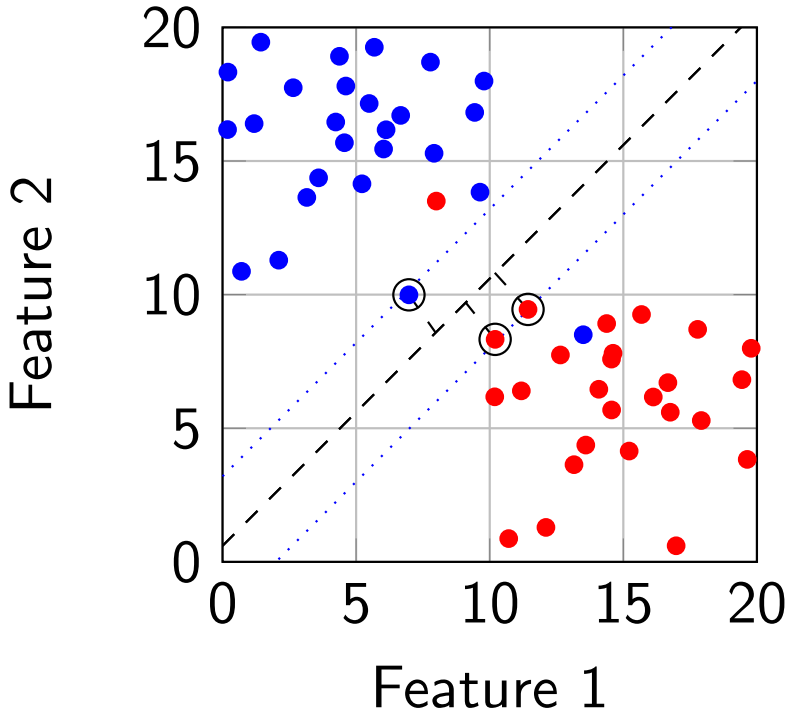

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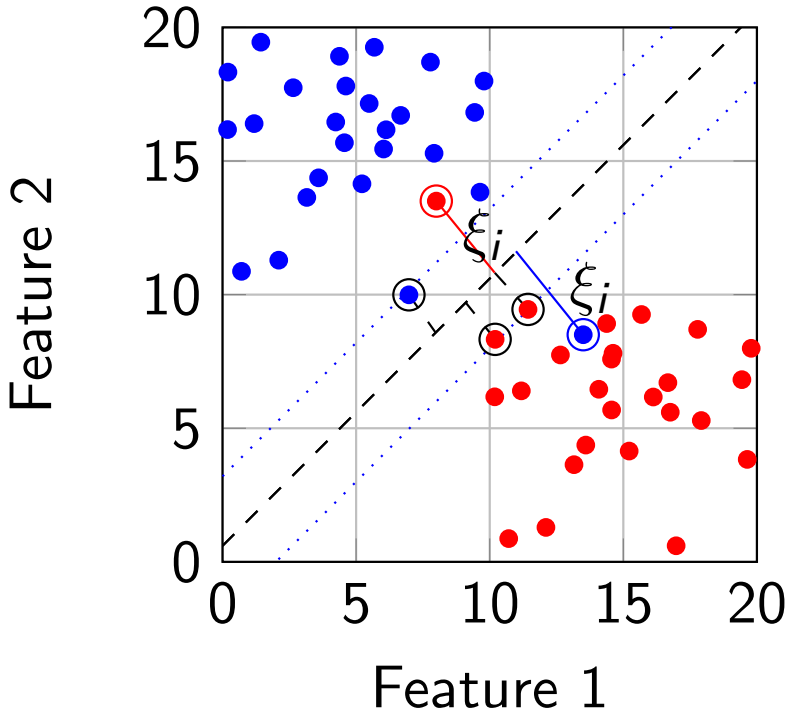

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Slack Variables

- Slack variables ξ_i are introduced for each data point.
- $\xi_i \geq 0$ measures the degree of misclassification of the i-th data point.
- If $\xi_i = 0$, the point is correctly classified and lies outside the margin.
- If $0 < \xi_i \leq 1$, the point is correctly classified but within the margin.
- If $\xi_i > 1$, the point is misclassified.

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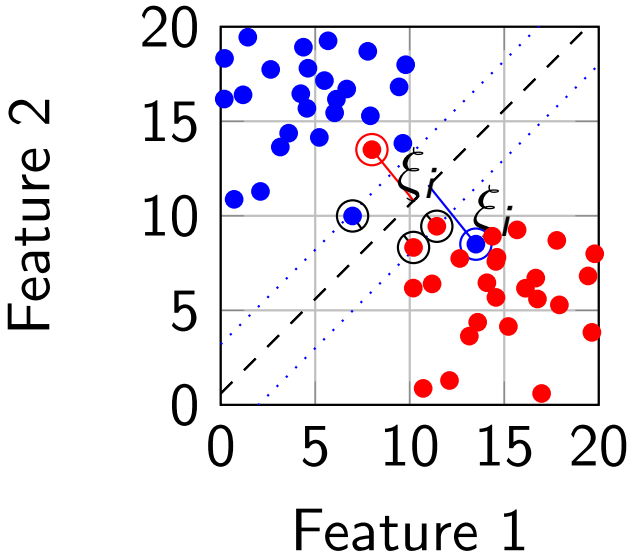

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Soft Margin SVM Optimization Problem

The optimization problem for Soft Margin SVM is:

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n} \xi_i$$

Subject to the constraints:

$$y_i(w^T x_i + b) \geq 1 - \xi_i \quad \forall i$$

- w: weight vector defining the hyperplane.
- b: bias term.
- ξ_i: slack variable allowing misclassifications.
- C: is a regularization parameter that controls the trade-off between maximizing the margin and minimizing the classification error.

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Linear SVMs: Overview

- The classifier is a separating hyperplane.
- Maximum margin classification ensures better generalization.
- Most "important" training points are support vectors; they define the hyperplane.
- Quadratic optimization algorithms can identify which training points xi are support vectors with non-zero Lagrangian multipliers.

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Non-Linear SVM

- Non-Linear SVM is used when data is not linearly separable in the original feature space.
- The data is transformed to a higher-dimensional feature space where it becomes linearly separable.
- This transformation is achieved using kernel functions.

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Non-linear SVMs

- Datasets that are linearly separable with some noise work out great.
- But what are we going to do if the dataset is just too hard?
- How about mapping data to a higher-dimensional space:
  - x_0
  - x_0
  - x_2
  - x_0

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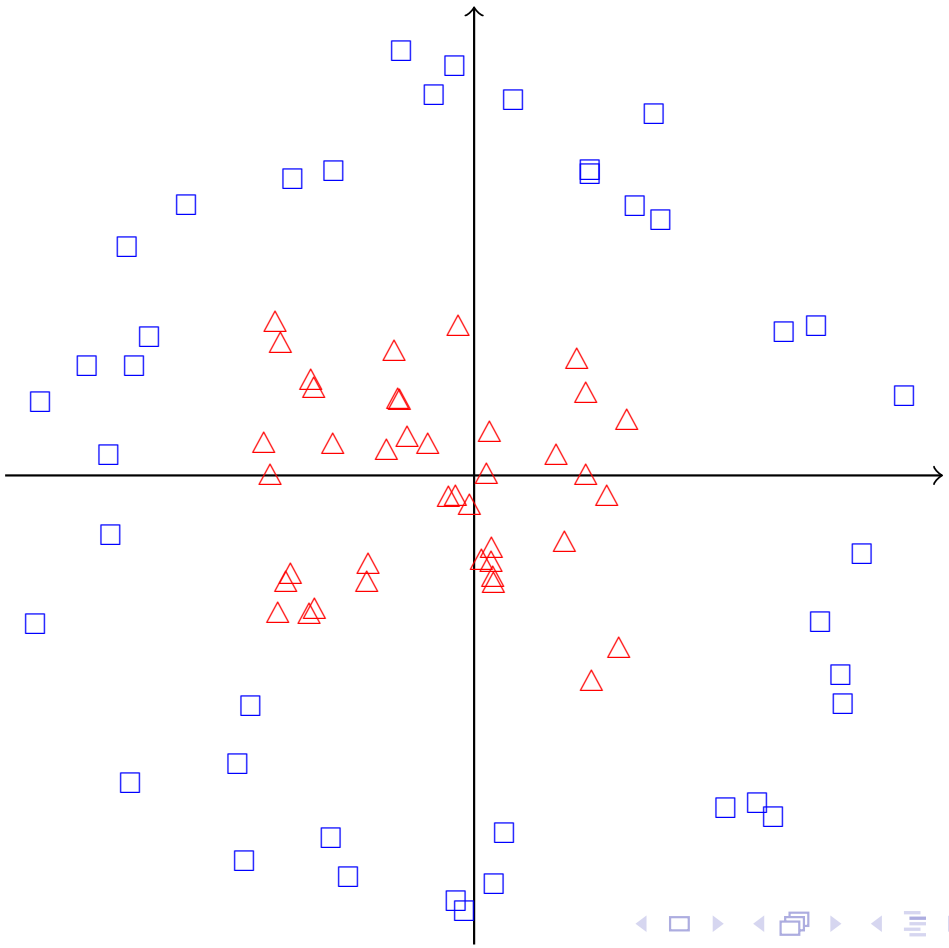
Non-linear SVMs: Feature Spaces

- General idea: the original feature space can always be mapped to some higher-dimensional feature space where the training set is separable:

$$\Phi : x \to \phi(x)$$



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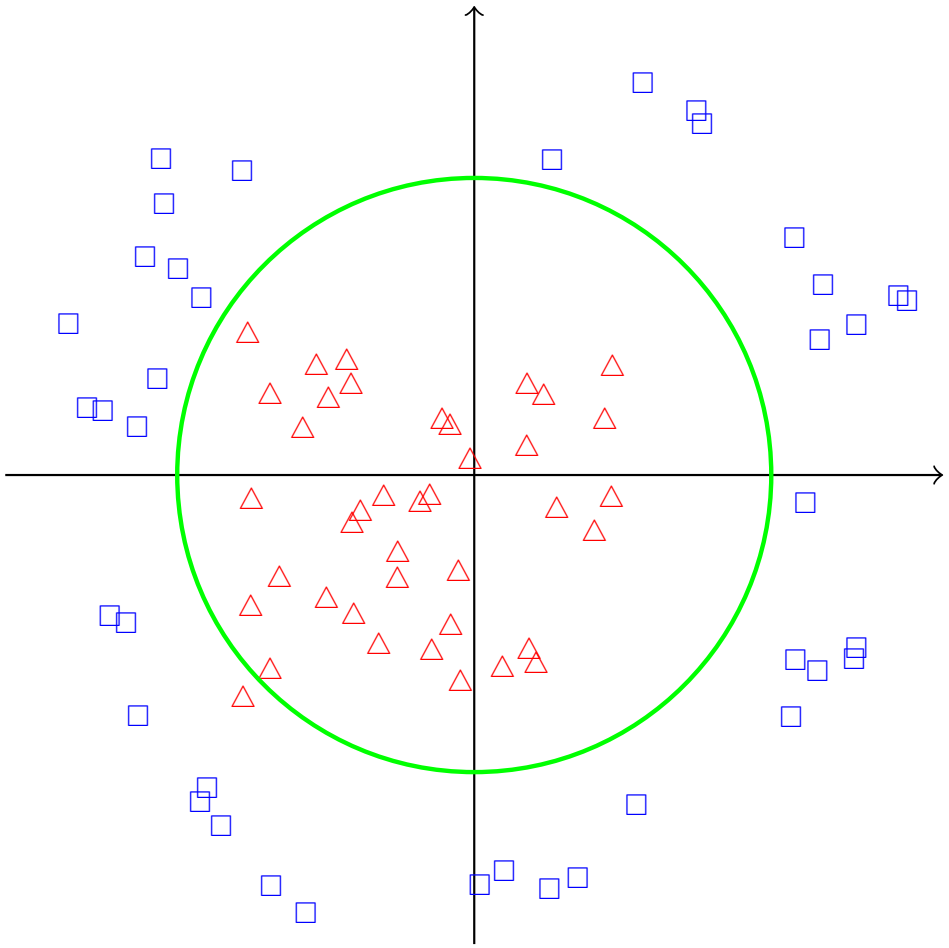
Kernel Trick

$$\Phi : x \rightarrow \phi(x)$$



(a) Input space

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Kernel Trick

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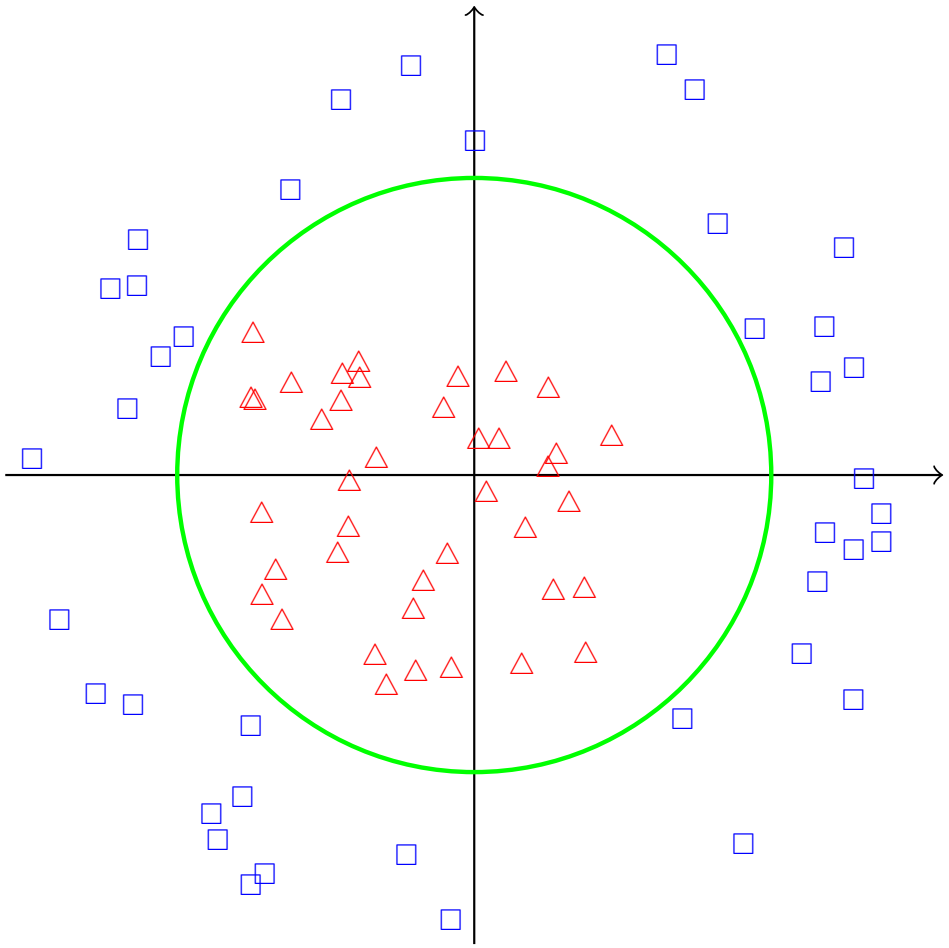

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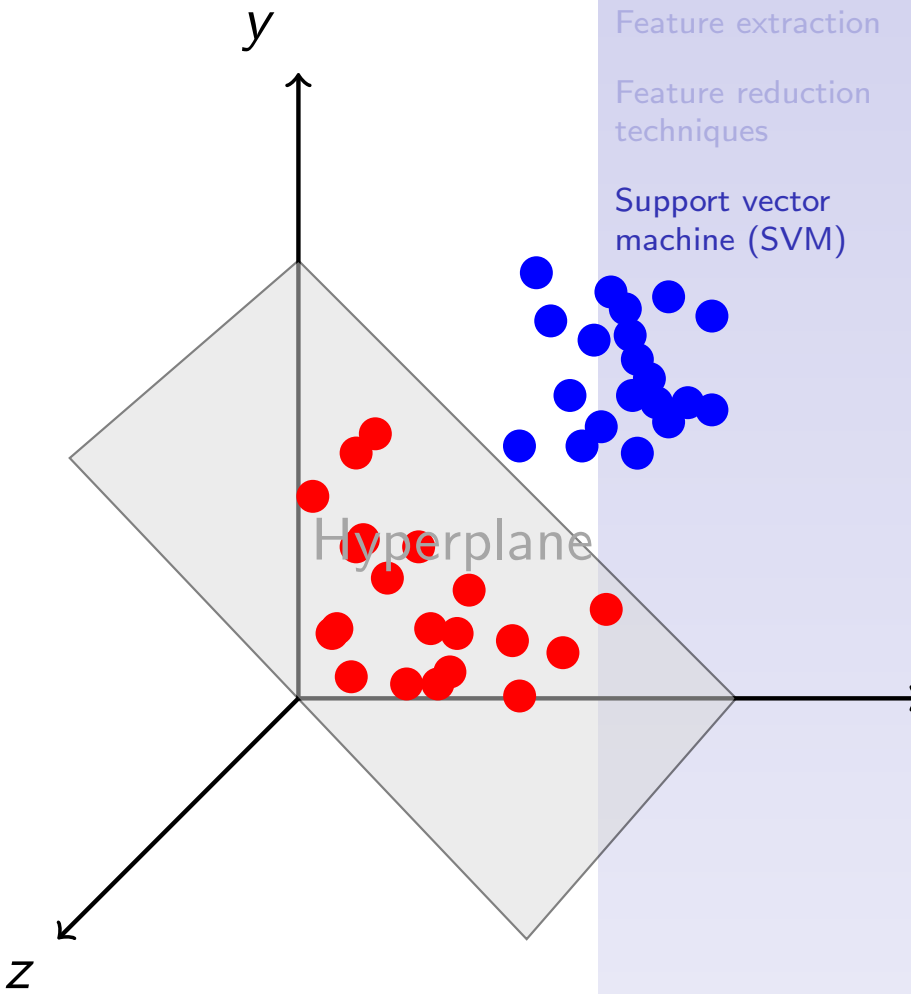

(b) Feature space

The Kernel Trick

- The linear classifier relies on inner product between vectors $K(x_i, x_j) = x_i^T x_j$.
- If every datapoint is mapped into high-dimensional space via some transformation $\Phi : x \to \phi(x)$, the inner product becomes:

$$K(x_i, x_j) = \phi(x_i)^T \phi(x_j)$$

- A function that takes as its inputs vectors in the original space and returns the dot product of the vectors in the feature space is called a **kernel function**.
- More formally, if we have data $\mathbf{x}, \mathbf{z} \in X$ and a map $\phi : X \to \mathbb{R}^N$, then:

$$k(\mathbf{x}, \mathbf{z}) = \langle \phi(\mathbf{x}), \phi(\mathbf{z}) \rangle$$

  is a **kernel function**.
- A kernel function is a function that is equivalent to an inner product in some feature space.

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Optimization in SVM

- Objective: Maximize the margin while minimizing classification errors.
- Primal optimization problem:

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi_i$$

$$\text{subject to } y_i(w \cdot x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

- Dual problem uses Lagrange multipliers for efficient computation.

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Solving the Optimization Problem

- Quadratic programming (QP) methods are used to solve the dual problem.
- Dual formulation:

$$\max_{\alpha} \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j K(x_i, x_j)$$

$$\text{subject to } \sum_{i=1}^{n} \alpha_i y_i = 0, \quad 0 \leq \alpha_i \leq C$$

- The solution provides the support vectors and decision boundary.

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Examples of Kernel Functions

- Linear: $K(x_i, x_j) = x_i^T x_j$
- Polynomial of power p: $K(x_i, x_j) = (1 + x_i^T x_j)^p$
- Gaussian (radial-basis function): $K(x_i, x_j) = \exp(-\|x_i - x_j\|^2/2\sigma^2)$
- Mapping $\Phi : x \to \phi(x)$, where $\phi(x)$ is infinite-dimensional: every point is mapped to a function (a Gaussian); combination of functions for support vectors is the separator.
- Higher-dimensional space still has intrinsic dimensionality d (the mapping is not onto), but linear separators in it correspond to non-linear separators in original space.

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Non-linear SVMs Mathematically

- Dual problem formulation: The solution is:

$$\max_{\alpha} Q(\alpha) = \sum \alpha_i - \frac{1}{2} \sum \sum \alpha_i \alpha_j y_i y_j K(x_i, x_j)$$

- Subject to:

$$\sum \alpha_i y_i = 0$$

$$\alpha_i \geq 0 \text{ for all } \alpha_i$$

- The optimization techniques for finding α_i's remain the same!
- The decision function is:

$$f(x) = \sum \alpha_i y_i K(x_i, x) + b$$

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Import SVM library and Implementation

```
# Import SVM and train test split
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
# Using 70% data for trainig and 30% data for testing
X_train, X_test, y_train, y_test = train_test_split(pca_result, Y,
     test_size=0.2, random_state=42)

# Initialize linear and nonlinear SVM models
linear_svm = SVC(kernel='linear')
nonlinear_svm = SVC(kernel='rbf')

# Fit the models with the selected features
linear_svm.fit(X_train, y_train)
nonlinear_svm.fit(X_train, y_train)

# Evaluate the models
linear_score = linear_svm.score(X_test, y_test)
nonlinear_score = nonlinear_svm.score(X_test, y_test)
```

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Plot SVM boundary

```
def plot_decision_boundary(clf, X, y, name):
# Convert X to NumPy array if it's a DataFrame
        if hasattr(X, "values"):
        X = X.values

# Create a mesh grid
        x_min, x_max = X[:, 0].min() - 1, X[:, 0].max() + 1
        y_min, y_max = X[:, 1].min() - 1, X[:, 1].max() + 1
        xx, yy = np.meshgrid(np.linspace(x_min, x_max, 500),
        np.linspace(y_min, y_max, 500))

# Predict on grid points
        Z = clf.decision_function(np.c_[xx.ravel(), yy.ravel()])
        Z = Z.reshape(xx.shape)

# Plot contours
        plt.contourf(xx, yy, Z, levels=np.linspace(Z.min(), Z.max(), 50)
              , cmap=plt.cm.coolwarm, alpha=0.8)
        plt.contour(xx, yy, Z, levels=[0], linewidths=2, colors='k')  #
              Decision boundary

# Scatter plot of data points
        plt.scatter(X[:, 0], X[:, 1], c=y, cmap=plt.cm.coolwarm,
              edgecolors='k')
        plt.xlabel('Feature 1')
        plt.ylabel('Feature 2')
        plt.title(name)
        plt.savefig(name)
        plt.show()


plot_decision_boundary(linear_svm, X_train, y_train,'SVM_Linear')
```


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Linear and Nonlinear SVM decision boundaries

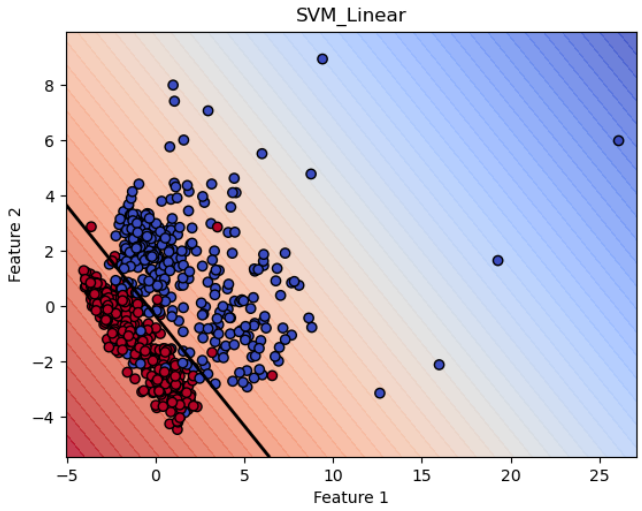

Figure: SVM linear classifier

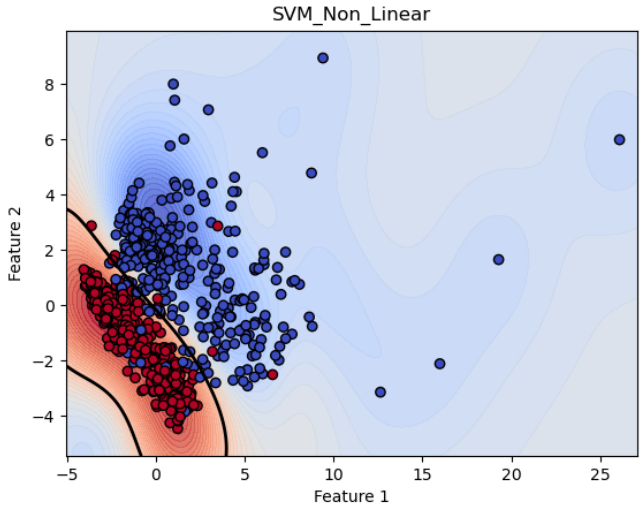

Figure: SVM nonlinear classifier (RBF kernel)

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Conclusion

- SVM is a powerful classifier for both linear and nonlinear problems.
- The choice of kernel function and the soft margin parameter C significantly impacts performance.
- Maximum margin classification ensures better generalization.

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